

Respostas dos exercícios

Capítulo I

1. São proposições: a, b, c, d, e, f, g.
São verdadeiras: a, c, d, e, g.

2. a) $3 \cdot 7 \neq 21$ (F)
b) $3(11 - 7) = 5$ (F)
c) $3 \cdot 2 + 1 \leq 4$ (F)
d) $5 \cdot 7 - 2 > 5 \cdot 6$ (V)
e) $\left(\frac{1}{2}\right)^7 \geq \left(\frac{1}{2}\right)^3$ (F)
f) $\sqrt{2} \geq 1$ (V)
g) $-(-4) < 7$ (V)
h) $3 \nmid 7$ (V)

3. a) V e) V
b) V f) F
c) V g) F
d) F

4. a) V e) F
b) V f) F
c) V g) V
d) V

5. a) F e) F
b) V f) V
c) V g) V
d) V h) V

6. p (V); q (V); r (F); s (F)

8. a) $(\exists x)(x^2 - 5x + 4 = 0)$
b) $(\forall a)[(a + 1)(a - 1) = a^2 - 1]$
c) $(\exists y)\left(\frac{y}{3} + \frac{y}{4} \neq \frac{y}{7}\right)$
d) $(\forall m)(\sqrt{m^2} + 9 \neq m + 3)$
e) $(\forall x)(-(-x) = x)$
f) $(\exists a)(5a + 4 \leq 11)$
g) $(\exists x)(\sqrt{x^2} = x)$
h) $(\exists a)\left(\frac{a^2 - a}{a} = a - 1\right)$

9. a) $\text{mdc}(2, 3) \neq 1$ e $\text{mmc}(2, 3) = 6$
b) $\frac{3}{5} \neq \frac{6}{10}$ e $3 \cdot 10 = 6 \cdot 5$
c) $\frac{3}{7} < 1$ ou $-3 < -7$
d) $2^2 = 4$ e $\sqrt{4} \neq 2$
e) $(-3)^2 = 9$ e $\sqrt{9} = -3$
f) $2 \leq 5$ e $3^2 > 5^2$
g) $(\exists x)(x > 2 \text{ e } 3^x \leq 3^2)$
h) $(\forall x)(\sqrt{x} \geq 0)$
i) Existe um número inteiro primo e par.
j) Existe um triângulo isósceles e não equilátero.
k) Todo losango é quadrado.
l) Todo número tem raiz quadrada diferente de zero.
m) Existe um triângulo equiângulo e não equilátero.

10. a) F d) F g) F j) V m) F
 b) F e) F h) V k) F
 c) V f) F i) V l) F

Capítulo II

13. a) $\{-9, -6, -3, 0, 3, 6, 9\}$
 b) $\{\pm 1, \pm 2, \pm 3, \pm 6, \pm 7, \pm 14, \pm 21, \pm 42\}$
 c) $\{0\}$
 d) $\left\{\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{2}{2}\right\}$
 e) $\{\text{Cuiabá, Campo Grande, Goiânia}\}$

14. $A = \{x \mid x \text{ é divisor de } 6\}$
 $B = \{x \mid x \text{ é múltiplo inteiro e negativo de } 10\}$
 $C = \{x \mid x \text{ é quadrado de um inteiro}\}$
 $D = \{x \mid x \text{ é satélite natural da Terra}\}$

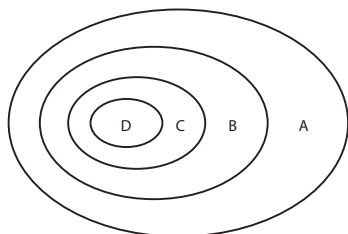
15. $D = \{3\}$

16. $B = \emptyset$

19. todas

20. a) V c) F e) F g) V i) V
 b) F d) F f) V h) V j) F

- 21.



22. $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, A\}$

23. $A \cup B = \{a, b, c, d\}$, $A \cup C = \{a, b, c, e\}$,
 $B \cup C = \{c, d, e\}$, $A \cup B \cup C = \{a, b, c, d, e\}$

25. a) V b) F c) F d) V e) V f) V

26. círculo de centro O e raio $2r$

27. plano α

28. $A \cap B = \{b, c, d\}$, $A \cap C = \{c\}$,
 $B \cap C = \{c, e\}$, $A \cap B \cap C = \{c\}$

30. a) V b) F c) F d) V e) V f) V

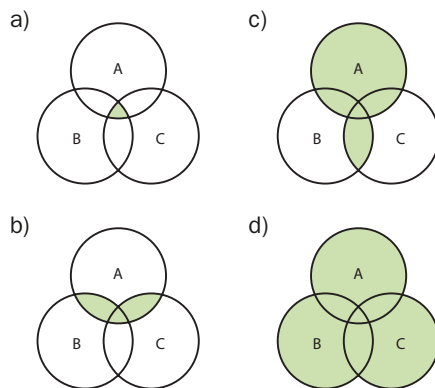
31. a) L b) R c) Q d) Q e) Q f) P

33. $X = \{a, c, e\}$

34. $C = \{2, 5, 6, 7, 9, 10\}$

35. 4: $\{1, 2\}$, $\{1, 2, 3\}$, $\{1, 2, 4\}$, $\{1, 2, 3, 4\}$

- 36.



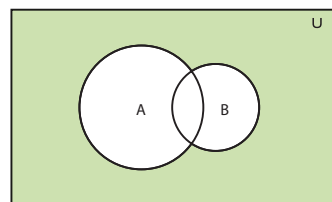
37. 2

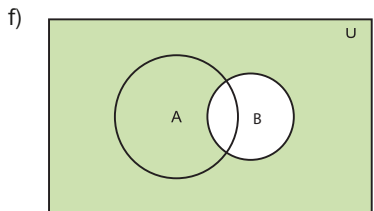
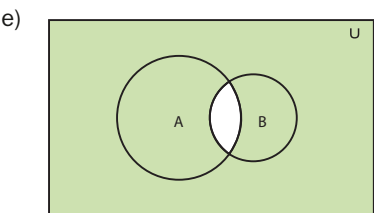
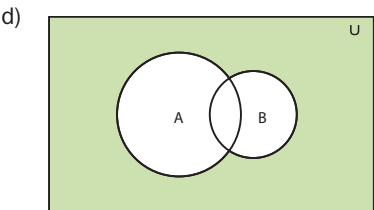
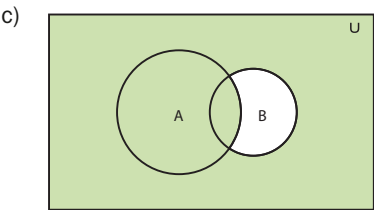
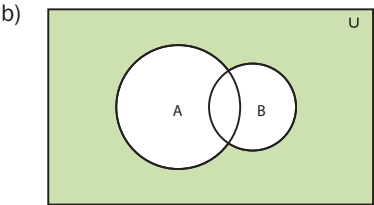
38. a) $\{a, b\}$ d) $\{a, b\}$
 b) $\{e, f, g\}$ e) $\{a, b, c\}$
 c) $\{b\}$ f) $\{a, c, e, f, g\}$

40. a) V b) V c) F d) V

41. $X = \{1, 3, 5\}$

42. a)





44. a) V b) V c) F d) V

45. $\bar{F} = \{6, 7, 8\}$

46. $A = \{6, -1\}$ $D = \left\{-\frac{1}{2}\right\}$

$B = \{e, x, r, c, i, o\}$ $E = \{2, 3, 4, 5\}$
 $C = \{3, -3, 5\}$

47. a, b, d, f

49. 64

50. $n_{A \cup B \cup C} = n_A + n_B + n_C - n_{A \cap B} - n_{B \cap C} -$
 $- n_{C \cap A} + n_{A \cap B \cap C}$

51. $n(A \cap B) = 8$

52. 332 e 83

53. $P' \cup Q$

54. a) 8 b) 1 c) 7 d) 3 e) 12

55. $n(A) = 4$; $n(B) = 4$

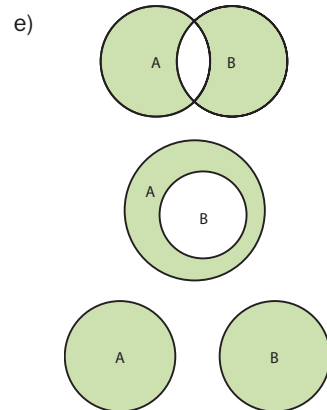
56. a) 500 b) 61 c) 257 d) 84

57. $A = \{p, q, r, s, t\}$
 $B = \{r, s, x, z\}$
 $C = \{s, t, u, v, x\}$

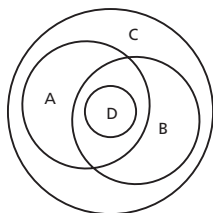
58. a) 560 b) 280

59. 40%

60. a) $\{a, b, e, f, g\}$



61.



Capítulo III

62. $n(H) = 14$

63. $n(X) = 22$

64. $B = \mathbb{N}^*$

65. a, c, d, g, h, i

66. $D(6) = \{\pm 1, \pm 2, \pm 3, \pm 6\}$
 $D(-18) = \{\pm 1, \pm 2, \pm 3, \pm 6, \pm 9, \pm 18\}$
 $D(-24) \cap D(16) = \{\pm 1, \pm 2, \pm 4, \pm 8\}$
 $M(4) = \{0, \pm 4, \pm 8, \pm 12, \dots\}$
 $M(10) = \{0, \pm 10, \pm 20, \pm 30, \dots\}$
 $M(-9) \cap M(6) = \{0, \pm 18, \pm 36, \pm 54, \dots\}$

67. $12, 0, -1, -4$ e 49

68. a) Não, pois
- $1 \in D(a) \cap D(b)$
- .
-
- b)
- m
- é um máximo divisor comum de
- a
- e
- b
- :
- $\text{mdc}(a, b) = \pm m$
- .
-
- c)
- a
- e
- b
- são primos entre si:
- $\text{mdc}(a, b) = \pm 1$
- .
-
- d) Quando
- a
- é múltiplo de
- b
- .
-
- e) Quando
- a
- e
- b
- são primos entre si.
-
- f)
- n
- é um mínimo múltiplo comum de
- a
- e
- b
- :
- $\text{mmc}(a, b) = \pm n$
- .

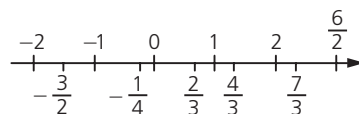
69. a) ± 1 d) ± 6
 b) ± 2 e) ± 12
 c) ± 2 f) ± 42

70. $a, b, c, d, e, f, h, k, l$

71. $\frac{2}{5}, \frac{4}{9}, \frac{8}{25}, \frac{32}{99}, \frac{271}{5}$ e $\frac{602}{111}$

72. $\frac{2}{3} < \frac{11}{12} < \frac{15}{16} < \frac{18}{19} < \frac{47}{48} < 1$

74.



75. a) 1 b) 2

76. 0,025

77. α é racional, $\alpha = 1,4111\dots = \frac{127}{90}$

78. 17 142,86 dólares

79. 20%

80. a, b, c, f, g, h, i

88. 1

89.



90. $[-1, 3] = \{x \in \mathbb{R} \mid -1 \leq x \leq 3\}$

$[0, 2[= \{x \in \mathbb{R} \mid 0 \leq x < 2\}$

$] -3, 4[= \{x \in \mathbb{R} \mid -3 < x < 4\}$

$] -\infty, 5[= \{x \in \mathbb{R} \mid x < 5\}$

$[1, +\infty[= \{x \in \mathbb{R} \mid x \geq 1\}$

92. a) $[1, 2]$ d) $[0, 2]$

b) $]1, 2[$ e) $[-1, 2[$

c) $]0, \frac{2}{5}[$ f) $[1, 2]$

93. a) $[-1, 4]$ c) $[-1, 5]$

b) $] -2, 5[$ d) $[-\frac{3}{2}, 0]$

94. $\mathbb{C}_A^B = [0, 1] \cup [3, 5[$

95. $\{x \in \mathbb{R} \mid -1 < x \leq 5\}$

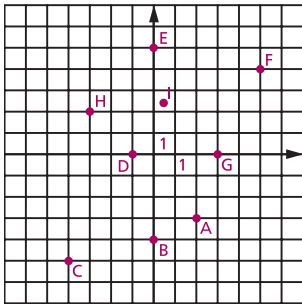
96. $[0, 2]$

97. $\mathbb{Z}$

Capítulo IV

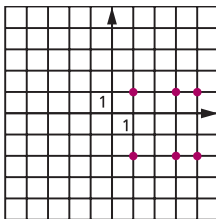
117. A(4, 2), B(-4, 6), C(-5, -3), D(4, -5),
E(0, 4), F(-3, 0), G(0, -6), H(5, 0), I(0, 0)

118.

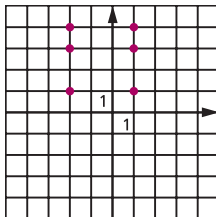


- 119.** a) $A \times B = \{(1, -2), (1, 1), (3, -2), (3, 1), (4, -2), (4, 1)\}$
b) $B \times A = \{(-2, 1), (-2, 3), (-2, 4), (1, 1), (1, 3), (1, 4)\}$
c) $A \times C = \{(1, -1), (1, 0), (1, 2), (3, -1), (3, 0), (3, 2), (4, -1), (4, 0), (4, 2)\}$
d) $C \times A = \{(-1, 1), (-1, 3), (-1, 4), (0, 1), (0, 3), (0, 4), (2, 1), (2, 3), (2, 4)\}$
e) $B^2 = \{(-2, -2), (-2, 1), (1, -2), (1, 1)\}$
f) $C^2 = \{(-1, -1), (-1, 0), (-1, 2), (0, -1), (0, 0), (0, 2), (2, -1), (2, 0), (2, 2)\}$

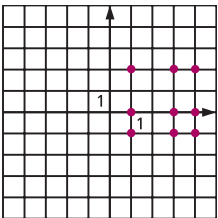
a)



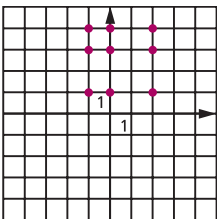
b)



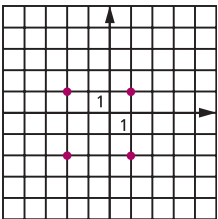
c)



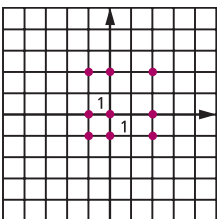
d)



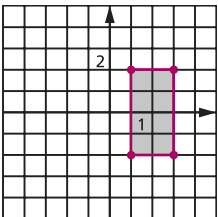
e)

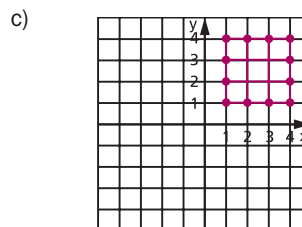
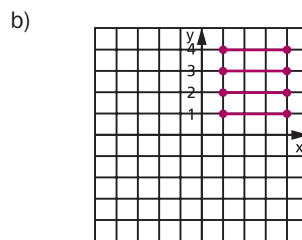
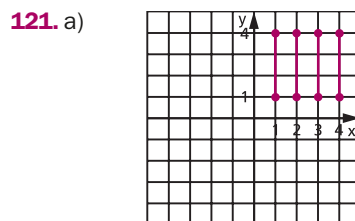
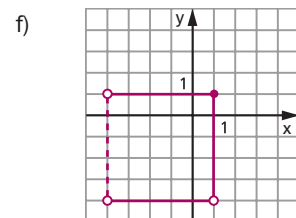
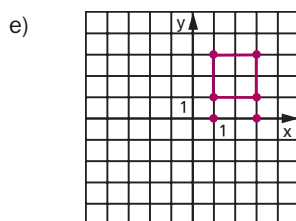
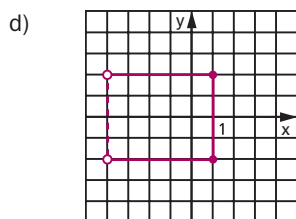
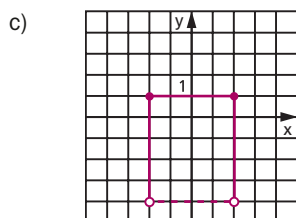
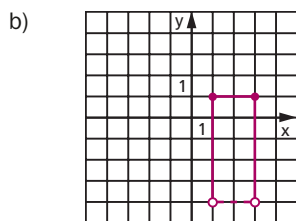


f)



120. a)





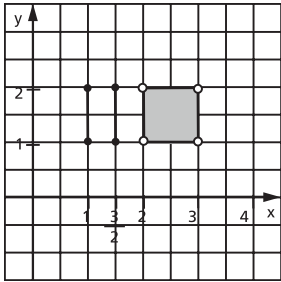
- 122.** $A \times A$ está contido em todos os nove.
 $A \times B$ está contido em $A \times B$, $A \times C$, $B \times B$, $B \times C$, $C \times C$ e $C \times B$.
 $A \times C$ está contido em $A \times C$, $B \times C$ e $C \times C$.
 $B \times A$ está contido em $A \times C$, $B \times A$, $B \times B$, $B \times C$, $C \times A$, $C \times B$ e $C \times C$.
 $C \times A$ está contido em $C \times A$, $C \times B$ e $C \times C$.
 $B \times B$ está contido em $B \times B$, $B \times C$, $C \times B$ e $C \times C$.
 $B \times C$ está contido em $B \times C$, $C \times C$ e $C \times B$.
 $C \times B$ está contido em $C \times B$ e $C \times C$.
 $C \times C$ está contido em $C \times C$.

124. $A^2 = \{(-2, -2), (-2, 0), (-2, 1), (-2, 3), (0, -2), (0, 0), (0, 1), (0, 3), (1, -2), (1, 0), (1, 1), (1, 3), (3, -2), (3, 0), (3, 1), (3, 3)\}$

125. $A \times B = \{(-1, -1), (-1, 0), (-1, 2), (-1, 5), (0, -1), (0, 0), (0, 2), (0, 5), (2, -1), (2, 0), (2, 2), (2, 5)\}$

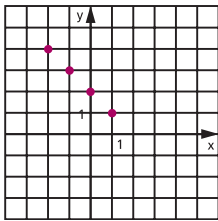
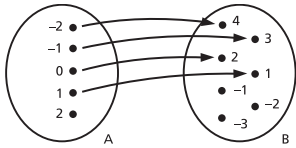
126. $n(F \times G) = 12$

127.

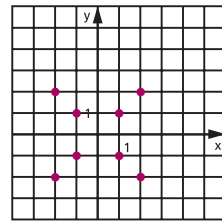
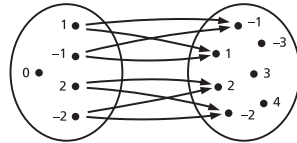


128. $n(D) = 3$

129. a) $R = \{(-2, 4), (-1, 3), (0, 2), (1, 1)\}$

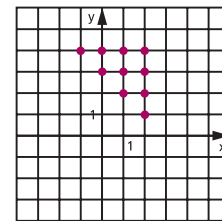
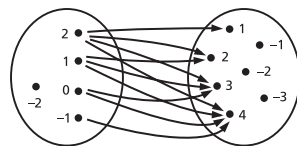
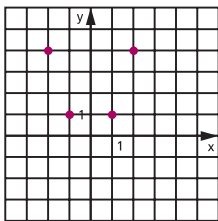
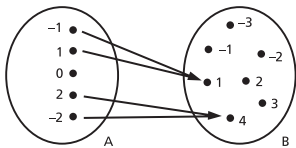


c) $T = \{(-2, -2), (-2, 2), (-1, -1), (-1, 1), (1, -1), (1, 1), (2, -2), (2, 2)\}$

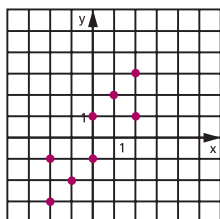
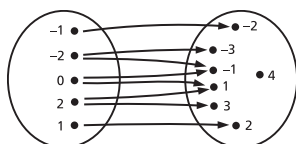


d) $V = \{(-1, 4), (0, 3), (0, 4), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4)\}$

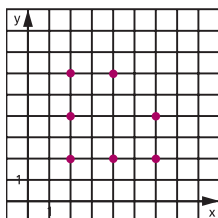
b) $S = \{(-2, 4), (2, 4), (-1, 1), (1, 1)\}$



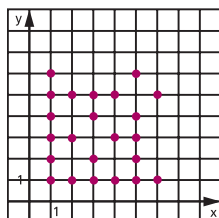
e) $W = \{(-2, -3), (-2, -1), (-1, -2), (0, -1), (0, 1), (1, 2), (2, 1), (2, 3)\}$



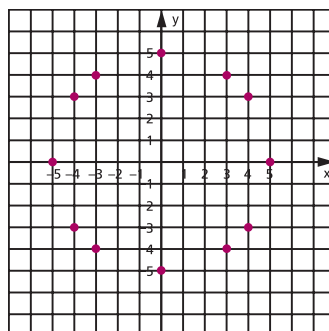
130. $R = \{(2, 2), (2, 4), (2, 6), (4, 2), (4, 6), (6, 2), (6, 4)\}$



131.



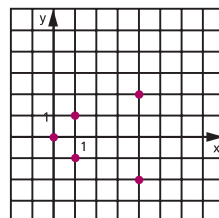
132.



133. a) $D = \{1, 2\}$ e $\text{Im} = \{1, 3, 4\}$
 b) $D = \{-2, -1, 3, 2\}$ e $\text{Im} = \{-7, 4, 1\}$
 c) $D = \{2, 1, 5\}$ e $\text{Im} = \{1, -3, \sqrt{2}\}$
 d) $D = \{1 + \sqrt{2}, 1 - \sqrt{3}\}$ e $\text{Im} = \{\sqrt{2}, 1\}$
 e) $D = \left\{3, \frac{5}{2}, \frac{3}{2}\right\}$ e $\text{Im} = \left\{\frac{1}{2}, -1, 0\right\}$

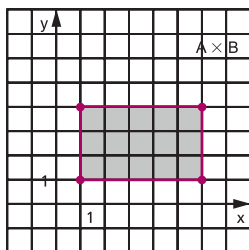
134. a) $D(R) = \{-2, -1, 0, 1\}$ e $\text{Im}(R) = \{1, 2, 3, 4\}$
 b) $D(S) = \{-2, -1, 1, 2\}$ e $\text{Im}(S) = \{1, 4\}$
 c) $D(T) = \{-2, -1, 1, 2\}$
 e $\text{Im}(T) = \{-2, -1, 1, 2\}$
 d) $D(V) = \{-1, 0, 1, 2\}$ e $\text{Im}(V) = \{1, 2, 3, 4\}$
 e) $D(W) = \{-2, -1, 0, 1, 2\}$
 e $\text{Im}(W) = \{-3, -2, -1, 1, 2, 3\}$

135. a) $R = \{(0, 0), (1, -1), (1, 1), (4, -2), (4, 2)\}$
 b) $D(R) = \{0, 1, 4\}$ e
 $\text{Im}(R) = \{-2, -1, 0, 1, 2\}$
 c)

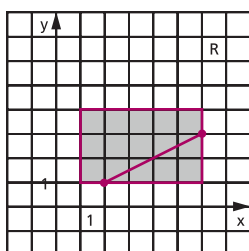


136. $D_f = \{x \in \mathbb{R} \mid x \neq -2 \text{ e } x \neq +2\}$

137. a)



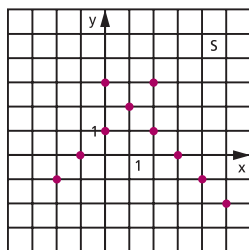
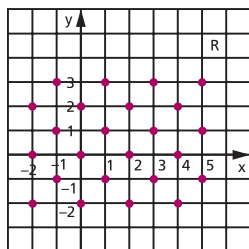
b)



c) $D(R) = \{x \in \mathbb{R} \mid 2 \leq x \leq 6\}$ e

$\text{Im}(R) = \{y \in \mathbb{R} \mid 1 \leq y \leq 3\}$

138. a)



b) $D_R = \{-2, -1, 0, 1, 2, 3, 4, 5\} = D_S$

$\text{Im}_R = \{-2, -1, 0, 1, 2, 3\} = \text{Im}_S$

c) $R \cap S = \emptyset$

139. a) $R^{-1} = \{(2, 1), (1, 3), (3, 2)\}$

b) $R^{-1} = \{(-1, 1), (-1, 2), (-1, 3), (1, -2)\}$

c) $R^{-1} = \{(-2, -3), (3, 1), (-3, 2), (1, 3)\}$

140. a) $R = R^{-1} = \{(0, 8), (1, 7), (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), (7, 1), (8, 0)\}$

b) $R = \{(0, 5), (2, 4), (4, 3), (6, 2), (8, 1), (10, 0)\}$

$R^{-1} = \{(5, 0), (4, 2), (3, 4), (2, 6), (1, 8), (0, 10)\}$

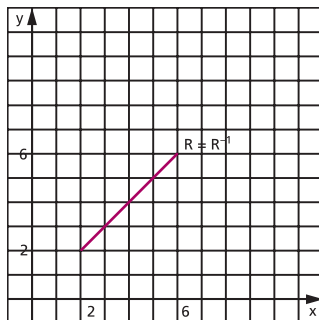
c) $R = \{(0, 10), (1, 5), (2, 2), (3, 1), (4, 2), (5, 5), (6, 10)\}$

$R^{-1} = \{(10, 0), (5, 1), (2, 2), (1, 3), (2, 4), (5, 5), (10, 6)\}$

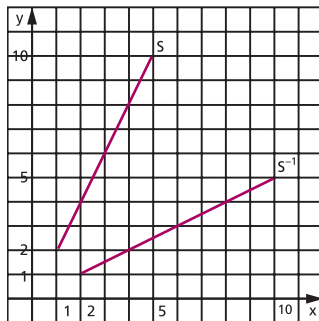
d) $R = \{(0, 1), (1, 2), (2, 4), (3, 8)\}$

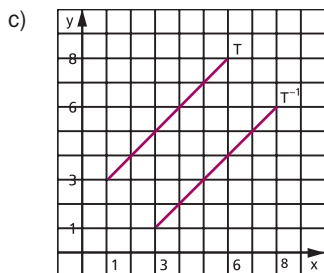
$R^{-1} = \{(1, 0), (2, 1), (4, 2), (8, 3)\}$

141. a)

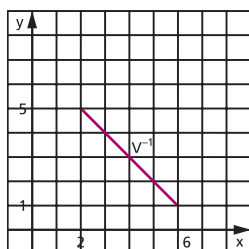
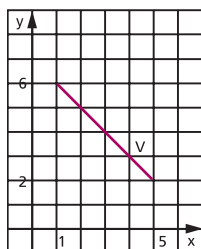


b)





d)



Capítulo V

142. a) Não define função de A em B, pois o elemento $2 \in A$ não está associado a nenhum elemento de B.
 b) Não define função de A em B, pois o elemento $1 \in A$ está associado a dois elementos de B.
 c e d) Define função de A em B, pois todo elemento de A está associado a um único elemento de B.

143. Somente d), pois o conjunto de partida é $A = \{0, 1, 2\}$ e o conjunto de chegada é $B = \{-1, 0, 1, 2\}$

144. a) É função.

b) Não é função de $\mathbb{R}$ em $\mathbb{R}$, pois qualquer reta vertical conduzida pelos pontos $(x, 0)$, com $x > 0$, encontra o gráfico da relação em dois pontos.

c) Não é função de $\mathbb{R}$ em $\mathbb{R}$, pois qualquer reta vertical conduzida pelos pontos $(x, 0)$, com $-1 < x < 1$, não encontra o gráfico da relação.

d) É função.

e) É função.

f) Não é função de $\mathbb{R}$ em $\mathbb{R}$, pois a reta vertical conduzida pelo ponto $(3, 0)$ encontra o gráfico da relação em mais que dois pontos e as retas verticais conduzidas pelos pontos $(x, 0)$, com $x \neq 3$, não encontram o gráfico da relação.

145. a) $f: \mathbb{R} \rightarrow \mathbb{R}$

$$x \mapsto -x$$

b) $g: \mathbb{R} \rightarrow \mathbb{R}$

$$x \mapsto x^3$$

c) $h: \mathbb{R} \rightarrow \mathbb{R}$

$$x \mapsto x^2 - 1$$

d) $k: \mathbb{R} \rightarrow \mathbb{R}$

$$x \mapsto 2$$

146. a) $f: \mathbb{Q} \rightarrow \mathbb{Q}$

$$x \mapsto -x + 1$$

c) $h: \mathbb{R}^* \rightarrow \mathbb{R}$

$$x \mapsto \frac{1}{x}$$

b) $g: \mathbb{Z} \rightarrow \mathbb{Q}$

$$x \mapsto 2^x$$

147. a) $f(2) = 4$

b) $f(-3) = -11$

c) $f(0) = -2$

d) $f\left(\frac{3}{2}\right)$ não tem significado, pois $\frac{3}{2} \notin \mathbb{Z}$.

148. a) $f(2) = 2$

b) $f(-1) = 8$

c) $f\left(\frac{1}{2}\right) = \frac{11}{4}$

d) $f\left(-\frac{1}{3}\right) = \frac{46}{9}$

e) $f(\sqrt{3}) = 7 - 3\sqrt{3}$

f) $f(1 - \sqrt{2}) = 4 + \sqrt{2}$

149. $f(2) = 17$

150. a) $f(3) = 1$

b) $f\left(-\frac{3}{7}\right) = 1$

c) $f(\sqrt{2}) = 1 + \sqrt{2}$

d) $f(\sqrt{4}) = 1$

e) $f(\sqrt{3} - 1) = \sqrt{3}$

f) $f(0,75) = 1$

152. $x = -4$

153. $x = 2$ ou $x = 3$

155. $f(0) = 0$ para $m \neq 0$

156. 32

158. a) $D(f) = \{0, 1, 2\}$ e $\text{Im}(f) = \{-1, 0, 1\}$

b) $D(g) = \{-1, 0, 1, 2\}$ e $\text{Im}(g) = \{1, 2\}$

c) $D(h) = \{-1, 0, 1\}$ e $\text{Im}(h) = \{-2\}$

d) $D(k) = \{-2, 0, 1, 2\}$ e
 $\text{Im}(k) = \{-2, -1, 0, 2\}$

159. a) $\text{Im} = \{-2, 0, 2\}$

b) $\text{Im} = \{y \in \mathbb{R} \mid -2 \leq y \leq 2\}$

c) $\text{Im} = \{y \in \mathbb{R} \mid y = 1 \text{ ou } y \geq 2\}$

d) $\text{Im} = \mathbb{R}$

e) $\text{Im} = \{y \in \mathbb{R} \mid 0 \leq y \leq 2 \text{ ou } y > 4\}$

f) $\text{Im} = \{y \in \mathbb{R} \mid y \leq 1\}$

160. a) $D = \{-4, -3, -2, -1, 0, 1, 2, 3\}$ e
 $\text{Im} = \{1, 2, 3, 4, 5\}$

b) $D = \{x \in \mathbb{R} \mid -2 \leq x \leq 3\}$ e
 $\text{Im} = \{y \in \mathbb{R} \mid -3 \leq y \leq 2\}$

c) $D = \{x \in \mathbb{R} \mid -2 \leq x \leq 4\}$ e
 $\text{Im} = \{y \in \mathbb{R} \mid 1 \leq y \leq 5\}$

d) $D = \{x \in \mathbb{R} \mid -3 \leq x < 5\}$ e
 $\text{Im} = \{y \in \mathbb{R} \mid 1 \leq y < 3\}$

e) $D = \{x \in \mathbb{R} \mid -4 \leq x \leq 4\}$ e
 $\text{Im} = \{y \in \mathbb{R} \mid -3 \leq y \leq 5\}$

f) $D = \{x \in \mathbb{R} \mid -3 \leq x < 4\}$ e
 $\text{Im} = \{-3, -2, -1, 0, 1, 2, 3\}$

161. a) $D(f) = \mathbb{R}$

b) $D(g) = \mathbb{R} - \{-2\}$

c) $D(h) = \mathbb{R} - \{2, -2\}$

d) $D(p) = \{x \in \mathbb{R} \mid x \geq 1\}$

e) $D(q) = \{x \in \mathbb{R} \mid x > -1\}$

f) $D(r) = \{x \in \mathbb{R} \mid x \geq -2 \text{ e } x \neq 2\}$

g) $D(s) = \mathbb{R}$

h) $D(t) = \mathbb{R} - \left\{-\frac{3}{2}\right\}$

i) $D(n) = \mathbb{R} - \{3\}$

162. $\{y \in \mathbb{R} \mid y \geq 2\}$

163. $[2, 6]$

164. Todas são iguais, pois são todas funções de $\mathbb{R}$ em $\mathbb{R}$ e associam cada número real ao seu cubo.

165. Não são iguais, pois para $x < 0$ temos $\sqrt{x^2} \neq x$.

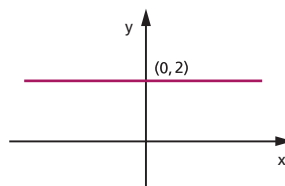
166. Somente serão iguais se forem funções de A em $\mathbb{R}$, em que A é qualquer subconjunto de $\{x \in \mathbb{R} \mid x \geq 1\}$.

167. São iguais, pois $\sqrt{\frac{x+1}{x^2-x}} = \sqrt{\frac{x+1}{x^2-x}}$ para $-1 \leq x < 0$ ou $x > 1$.

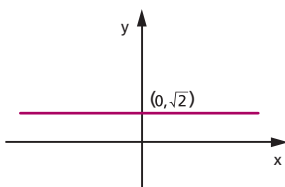
168. Não são iguais, pois não têm o mesmo domínio.

Capítulo VI

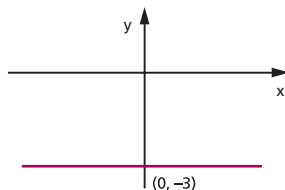
169. a)

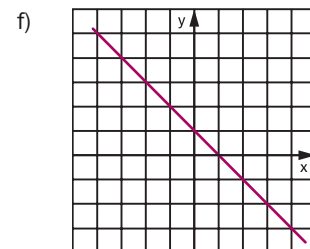
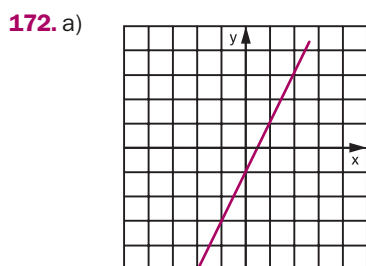
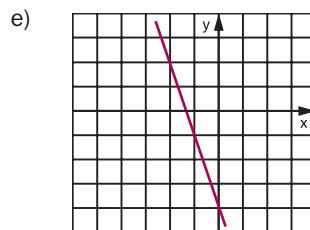
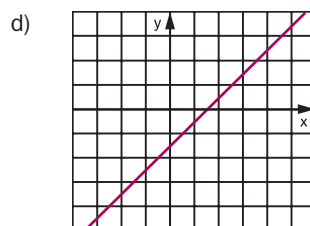
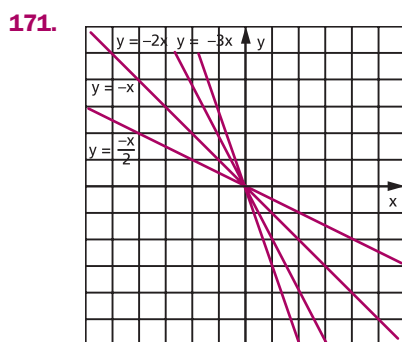
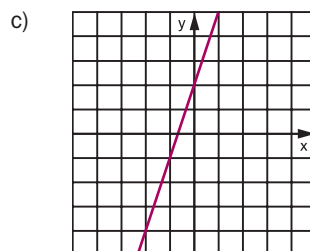
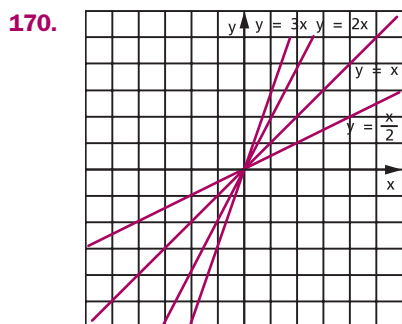
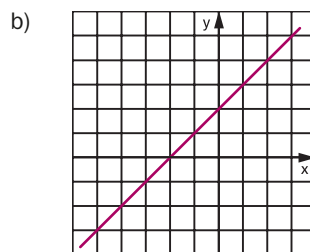
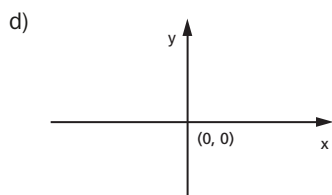


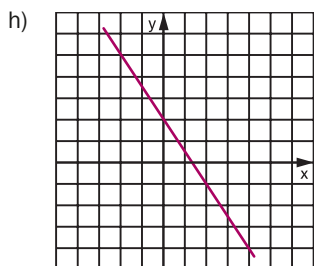
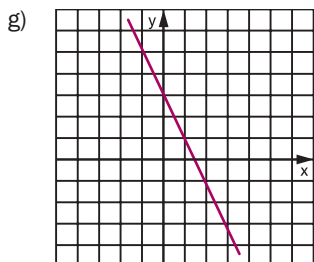
b)



c)







174. a) $S = \{(3, 2)\}$ d) $S = \{(3, -2)\}$

b) $S = \{(-2, 4)\}$ e) $S = \emptyset$

c) $S = \{(2, -1)\}$ f) $S = \{(0, 0)\}$

175. a) $S = \{(3, -1)\}$ b) $S = \{(2, 1)\}$

177. a) $y = 2x - 1$ c) $y = x - 5$

b) $y = \frac{1 - 3x}{2}$ d) $y = 2$

178. 23 brancas; 16 pretas

179. $f(3) = -1$

181. $y = -3x - 2$

182. $y = -\frac{x}{2} - \frac{1}{2}$

183. $y = \frac{3}{2}x + 4$

184. $y = -\frac{x}{3} - 3$

185. a) $y = \frac{x}{3} + \frac{1}{3}$ c) $y = \frac{2x}{3} - \frac{1}{3}$

b) $y = -\frac{x}{2} + 4$ d) $y = 2x + 3$

186. 20 litros

187. a) $x \leq 1637,11 \Rightarrow f(x) = 0$

$1637,12 \leq x \leq 2453,50 \Rightarrow$

$\Rightarrow f(x) = \frac{3x}{40} - 122,78$

$2453,51 \leq x \leq 3271,38 \Rightarrow$

$\Rightarrow f(x) = \frac{3x}{20} - 306,80$

$3271,39 \leq x \leq 4087,65 \Rightarrow$

$\Rightarrow f(x) = \frac{9x}{40} - 552,15$

$x > 4087,65 \Rightarrow f(x) = \frac{11x}{40} - n$

b) $n = 756,53$

188. mãe: $\frac{H}{8}$; cada menino: $\frac{H}{4}$; a menina: $\frac{3H}{8}$

189. 3300 km

190. R\$ 100,00

191. R\$ 8,00

192. 1

193. 25

194. a) crescente para $x \in \mathbb{R} \mid x \leq -2$ ou $x \geq 1$

decrescente para $x \in \mathbb{R} \mid -2 \leq x \leq 1$

b) crescente para $x \in \mathbb{R} \mid -1 \leq x \leq 0$ ou $x \geq 1$

decrescente para $x \in \mathbb{R} \mid x \leq -1$ ou $0 \leq x \leq 1$

c) crescente para $x \in \mathbb{R} \mid x \leq 0$ ou $x > 0$

196. a) crescente d) decrescente

b) decrescente e) decrescente

c) crescente f) crescente

198. a) crescente para $m > -2$
decrescente para $m < -2$
constante para $m = -2$

b) crescente para $m < 4$
decrescente para $m > 4$
constante para $m = 4$

c) crescente para $m < -3$
decrescente para $m > -3$
constante para $m = -3$

d) crescente para $m > 1$
decrescente para $m < 1$
constante para $m = 1$

199. a) $f(x) = 0 \Leftrightarrow x = -5$ ou $x = -3$ ou $x = 2$
ou $x = 6$

$f(x) > 0 \Leftrightarrow x < -5$ ou $-3 < x < 2$ ou $x > 6$

$f(x) < 0 \Rightarrow -5 < x < -3$ ou $2 < x < 6$

b) $g(x) = 0 \Leftrightarrow x = -3$ ou $x = -1$ ou $x = 3$

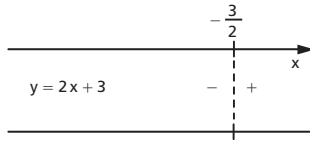
$g(x) > 0 \Leftrightarrow -3 < x < -1$

$g(x) < 0 \Leftrightarrow x < -3$ ou $x > -1$ e $x \neq 3$

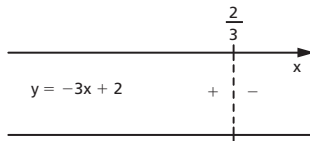
c) $h(x) = 0 \Leftrightarrow x = -2$

$h(x) > 0 \Leftrightarrow x \neq -2$

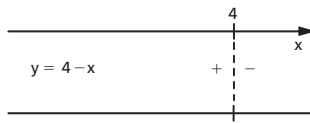
200. a)



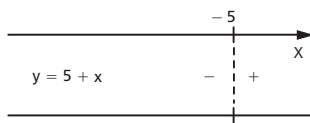
b)



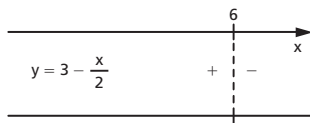
c)



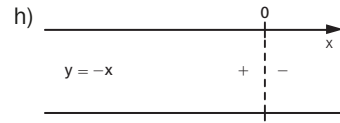
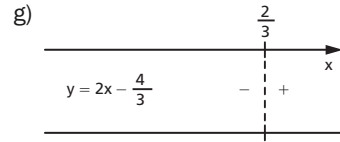
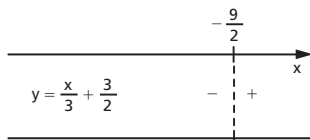
d)



e)



f)



202. $x < 3$

203. $x > \frac{4}{3}$

204. a) $x \geq -\frac{1}{5}$ b) $x > \frac{1}{2}$ c) $\forall x \in \mathbb{R}$

205. a) $x > 2$

b) $x \geq 0$

c) $\nexists x \in \mathbb{R}$

d) $x < -2$

e) $x \leq 3$

206. a) $S = \{x \in \mathbb{R} \mid x > -4\}$

b) $S = \{x \in \mathbb{R} \mid x \leq -10\}$

c) $S = \left\{x \in \mathbb{R} \mid x \geq -\frac{3}{4}\right\}$

208. a) $S = \{x \in \mathbb{R} \mid x \geq 3\}$

b) $S = \{x \in \mathbb{R} \mid x > -3\}$

c) $S = \{x \in \mathbb{R} \mid x \geq 7\}$

d) $S = \{x \in \mathbb{R} \mid x < 0\}$

e) $S = \emptyset$

f) $S = \mathbb{R}$

209. 7,9 ou mais.

211. a) $S = \{x \in \mathbb{R} \mid x > 1\}$

b) $S = \left\{x \in \mathbb{R} \mid x < \frac{1}{2}\right\}$

c) $S = \left\{x \in \mathbb{R} \mid x > -\frac{2}{3}\right\}$

212. a) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{3} < x < \frac{5}{3}\right\}$

b) $S = \left\{x \in \mathbb{R} \mid \frac{1}{2} \leq x < 4\right\}$

c) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{3} < x < 1\right\}$

d) $S = \emptyset$

e) $S = \left\{x \in \mathbb{R} \mid x < \frac{1}{3}\right\}$

f) $S = \{x \in \mathbb{R} \mid x > 1\}$

213. a) $S = \{x \in \mathbb{R} \mid 1 \leq x \leq 2\}$

b) $S = \{x \in \mathbb{R} \mid x < -3\}$

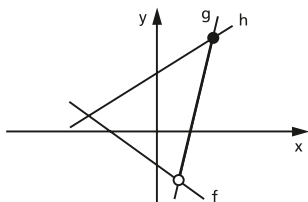
c) $S = \{x \in \mathbb{R} \mid 3 \leq x \leq 6\}$

d) $S = \emptyset$

e) $S = \left\{x \in \mathbb{R} \mid -1 < x < \frac{2}{9}\right\}$

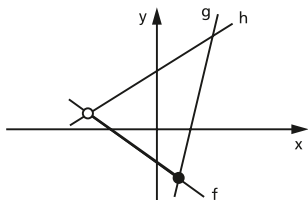
f) $S = \{x \in \mathbb{R} \mid -1 < x < 1\}$

214. a)



$S = \{x \in \mathbb{R} \mid 1 < x \leq 4\}$

b)



$S = \{x \in \mathbb{R} \mid -3 < x \leq 1\}$

c) $S = \emptyset$

215. a) $S = \left\{x \in \mathbb{R} \mid x < -1 \text{ ou } x > \frac{3}{5}\right\}$

b) $S = \left\{x \in \mathbb{R} \mid x < -\frac{5}{2} \text{ ou } x > 2\right\}$

c) $S = \left\{x \in \mathbb{R} \mid x < -\frac{3}{4} \text{ ou } -\frac{2}{5} < x < 2\right\}$

d) $S = \left\{x \in \mathbb{R} \mid -\frac{2}{3} < x < \frac{4}{3} \text{ ou } x > 6\right\}$

e) $S = \left\{x \in \mathbb{R} \mid x \leq -\frac{7}{2} \text{ ou } x \geq \frac{1}{6}\right\}$

f) $S = \left\{x \in \mathbb{R} \mid -\frac{2}{7} \leq x \leq \frac{5}{2}\right\}$

g) $S = \left\{x \in \mathbb{R} \mid x \leq -\frac{3}{5} \text{ ou } -\frac{1}{4} \leq x \leq \frac{3}{2}\right\}$

h) $S = \left\{x \in \mathbb{R} \mid \frac{1}{4} \leq x \leq \frac{5}{3} \text{ ou } x \geq \frac{7}{2}\right\}$

216. a) $S = \{x \in \mathbb{R} \mid x \neq 3\}$

b) $S = \left\{x \in \mathbb{R} \mid x < -\frac{8}{3}\right\}$

c) $S = \emptyset$

d) $S = \left\{x \in \mathbb{R} \mid x < \frac{1}{7}\right\}$

e) $S = \mathbb{R}$

f) $S = \left\{x \in \mathbb{R} \mid x \leq -\frac{1}{5}\right\}$

g) $S = \left\{-\frac{4}{3}\right\}$

h) $S = \left\{x \in \mathbb{R} \mid x \geq \frac{8}{3}\right\}$

218. a) $S = \left\{x \in \mathbb{R} \mid x \geq \frac{2}{7}\right\}$

b) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{3} < x < \frac{2}{5}\right\}$

c) $S = \left\{x \in \mathbb{R} \mid x \leq -6 \text{ ou } x = \frac{1}{3} \text{ ou } x = -\frac{5}{4}\right\}$

d) $S = \left\{x \in \mathbb{R} \mid x \geq \frac{1}{5} \text{ ou } x = -3\right\}$

219. $S = \left\{x \in \mathbb{R} \mid x < 0 \text{ ou } \frac{2}{3} < x < 2\right\}$

220. a) $S = \left\{x \in \mathbb{R} \mid x < -2 \text{ ou } x > -\frac{1}{2}\right\}$

b) $S = \left\{x \in \mathbb{R} \mid x < \frac{2}{3} \text{ ou } x > \frac{3}{2}\right\}$

$$c) S = \left\{ x \in \mathbb{R} \mid -\frac{1}{5} < x \leq \frac{3}{4} \right\}$$

$$d) S = \left\{ x \in \mathbb{R} \mid x \leq -\frac{3}{2} \text{ ou } x > -\frac{1}{3} \right\}$$

$$221. a) S = \left\{ x \in \mathbb{R} \mid x < \frac{7}{8} \text{ ou } x > \frac{4}{3} \right\}$$

$$b) S = \{ x \in \mathbb{R} \mid -2 \leq x < -1 \}$$

$$c) S = \{ x \in \mathbb{R} \mid -3 < x < 15 \}$$

$$d) S = \left\{ x \in \mathbb{R} \mid x < -10 \text{ ou } x > -\frac{4}{3} \right\}$$

$$e) S = \{ x \in \mathbb{R} \mid 1 \leq x < 2 \}$$

$$f) S = \{ x \in \mathbb{R} \mid 2 < x \leq 3 \}$$

$$222. a) S = \left\{ x \in \mathbb{R} \mid -\frac{3}{4} < x < \frac{1}{2} \text{ ou } x > 4 \right\}$$

$$b) S = \left\{ x \in \mathbb{R} \mid x < -\frac{5}{2} \text{ ou } -\frac{3}{5} < x < -\frac{1}{3} \right\}$$

$$c) S = \left\{ x \in \mathbb{R} \mid x \leq -\frac{4}{5} \text{ ou } -\frac{1}{4} \leq x < \frac{5}{4} \right\}$$

$$d) S = \left\{ x \in \mathbb{R} \mid \frac{1}{2} \leq x < 3 \text{ ou } x > 5 \right\}$$

$$223. a) S = \{ x \in \mathbb{R} \mid -3 < x < 4 \text{ ou } x > 11 \}$$

$$b) S = \{ x \in \mathbb{R} \mid 0 < x < 1 \text{ ou } x > 2 \}$$

$$c) S = \{ x \in \mathbb{R} \mid -4 < x < -2 \}$$

$$d) S = \left\{ x \in \mathbb{R} \mid x < -\frac{5}{3} \text{ ou } -\frac{29}{24} \leq x < -\frac{2}{3} \right\}$$

$$e) S = \left\{ x \in \mathbb{R} \mid -\frac{5}{4} < x < -\frac{9}{42} \text{ ou } x > \frac{1}{4} \right\}$$

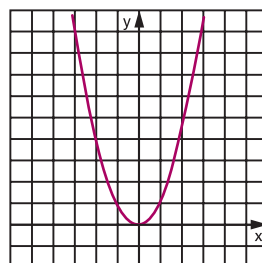
$$f) S = \left\{ x \in \mathbb{R} \mid x < 1 \text{ ou } \frac{3}{2} < x < 2 \right. \\ \left. \text{ou } x > 3 \right\}$$

$$g) S = \left\{ x \in \mathbb{R} \mid -1 < x \leq 0 \text{ ou } \frac{1}{3} < x < 1 \right. \\ \left. \text{ou } x \geq 3 \right\}$$

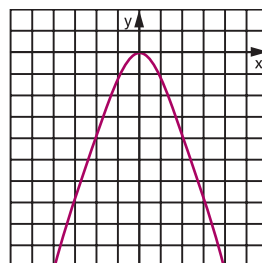
$$224. S = \{ x \in \mathbb{R} \mid x < 0 \text{ ou } x \geq 2 \}$$

Capítulo VII

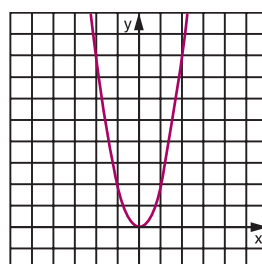
225. a)



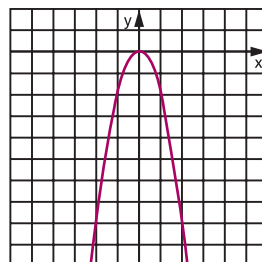
b)

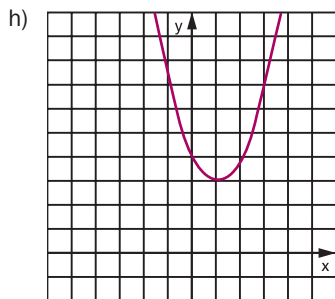
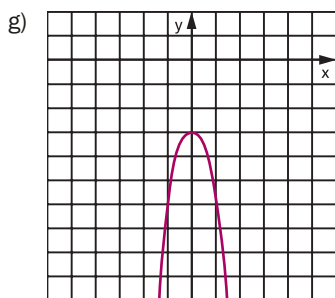
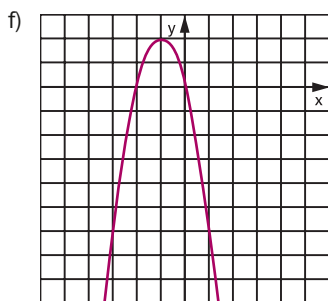
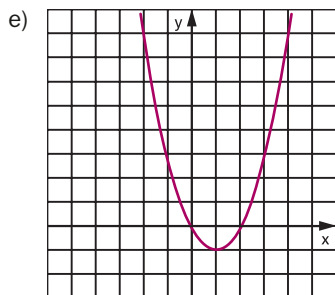


c)



d)





226. $m \neq 2$ e $m \neq -2$

227. $f(x) = -2x^2 + 3x + 1$

228. $abc = -70$

229. a) $x = 1$ ou $x = 2$

b) $x = 3$ ou $x = 4$

c) $x = 2$ ou $x = \frac{1}{3}$

d) Não existe $x \in \mathbb{R}$.

e) $x = -2$

f) $x = -\frac{1}{2}$ ou $x = 2$

g) $x = 1 + \sqrt{2}$ ou $x = 1 - \sqrt{2}$

h) Não existe $x \in \mathbb{R}$.

i) $x = \frac{\sqrt{2}}{2}$

j) $x = -1$ ou $x = \sqrt{3}$

k) $x = 0$ ou $x = 2$

l) $x = \sqrt{2}$ ou $x = -\sqrt{2}$

m) Não existe $x \in \mathbb{R}$.

n) $x = 0$

230. 50

231. $S = \{(3, 4), (4, 3)\}$

232. a) $S = \{-1, 4\}$

b) $S = \{(4, -4), (-1, 6)\}$

234. a) $x = 1$ ou $x = -1$ ou $x = 2$ ou $x = -2$

b) $x = 3$ ou $x = -3$

c) $x = \sqrt{3}$ ou $x = -\sqrt{3}$

d) $x = \sqrt{2}$ ou $x = -\sqrt{2}$

e) Não existe $x \in \mathbb{R}$.

f) Não existe $x \in \mathbb{R}$.

g) $x = 0$ ou $x = 2$ ou $x = -2$

h) $x = 2$ ou $x = -1$

236. $m > -\frac{9}{16}$ e $m \neq 1$

237. $m \leq \frac{17}{16}$ e $m \neq -2$

238. $m = -1$ ou $m = \frac{1}{3}$

239. $m = -2$ ou $m = \frac{2}{5}$

240. $m < -\frac{13}{12}$

241. $m < -\frac{1}{4}$

242. São as mesmas de $ax^2 + bx + c$, multiplicadas por $\alpha\beta$.

244. a) $\frac{5}{2}$

d) $\frac{29}{4}$

b) $-\frac{1}{2}$

e) $-\frac{29}{2}$

c) -5

f) $\frac{155}{8}$

245. $m = 2\sqrt{2}$

246. $|x_1 - x_2| = 46$

247. $\frac{b^2 - 2ac}{c^2}$

248. $m = -3$

249. $k = 6$

251. a) $x^2 + x - 6 = 0$

b) $4x^2 + 4x - 3 = 0$

c) $x^2 - 5,4x + 2 = 0$

d) $x^2 - (1 - \sqrt{2})x - \sqrt{2} = 0$

e) $x^2 - 2x - 2 = 0$

252. a) $a^2x^2 - (b^2 - 2ac)x + c^2 = 0$

b) $cx^2 + bx + a = 0$

c) $acx^2 - (b^2 - 2ac)x + ac = 0$

d) $a^3x^2 + (b^3 - 3abc)x + c^3 = 0$

253. $m = -2 + \sqrt{6}$ ou $m = -2 - \sqrt{6}$

254. $g(x) = x^2 - \frac{p}{q}x + \frac{1}{q}$

255. $m + n = 80$

256. a) $V(0, -4)$

d) $V(\frac{1}{4}, \frac{25}{16})$

b) $V(\frac{3}{2}, \frac{9}{4})$

e) $V(\frac{1}{2}, \frac{1}{36})$

c) $V(\frac{5}{4}, -\frac{9}{8})$

f) $V(\frac{7}{6}, -\frac{121}{36})$

257. a) $x_m = -\frac{5}{4}$ e $y_m = -\frac{25}{8}$

b) $x_M = 2$ e $y_M = 12$

c) $x_m = 1$ e $y_m = 0$

d) $x_m = \frac{7}{4}$ e $y_m = -\frac{9}{16}$

e) $x_M = \frac{5}{2}$ e $y_M = -\frac{3}{4}$

f) $x_M = \frac{4}{3}$ e $y_M = \frac{7}{18}$

258. $m = 2$

259. $m = -2$ ou $m = 1$

260. $m = -1$

261. Não existe $m \in \mathbb{R}$.

263. $y_M = y_V = \frac{21}{4}$; $y_m = f(6) = -7$

264. Não tem máximo, porque $a > 0$.

265. $v = 8$

266. $x = 2$ e $z = 4$

267. quadrado de lado 5 cm

268. 3 e 3

269. retângulo de lados $\frac{5}{8}$ e $\frac{5}{2}$

270. retângulo de lados 4 cm e 3 cm

271. retângulo de lados 2 cm e $\sqrt{3}$ cm

272. retângulo de lados 2 cm e 3 cm

273. 4

274. 2

275. a) $Im = \{y \in \mathbb{R} \mid y \geq -\frac{9}{4}\}$

b) $Im = \{y \in \mathbb{R} \mid y \leq 4\}$

c) $Im = \{y \in \mathbb{R} \mid y \geq -\frac{3}{4}\}$

d) $Im = \{y \in \mathbb{R} \mid y \leq 16\}$

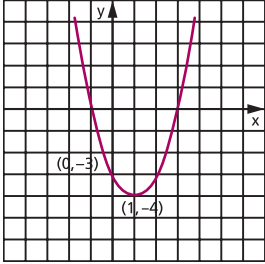
e) $Im = \{y \in \mathbb{R} \mid y \leq \frac{25}{16}\}$

f) $Im = \{y \in \mathbb{R} \mid y \geq \frac{1}{2}\}$

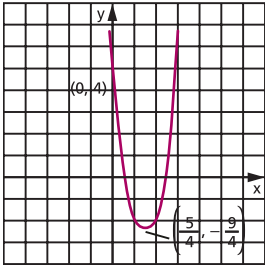
276. $m = \frac{10}{3}$

277. $m = \sqrt{10}$ ou $m = -\sqrt{10}$

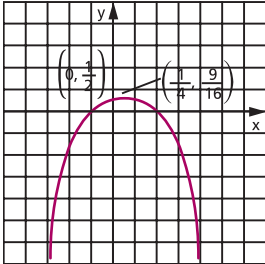
281. a)



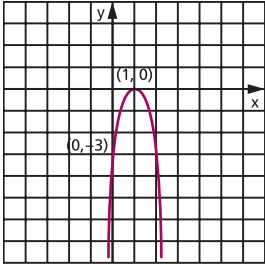
b)



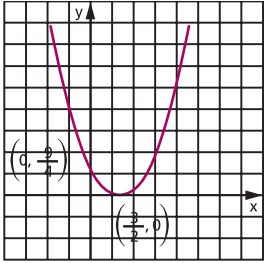
c)



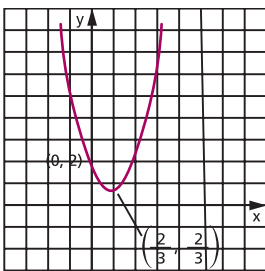
d)



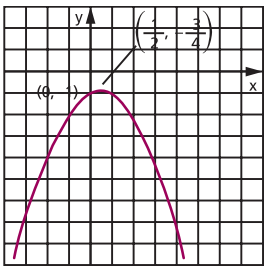
e)



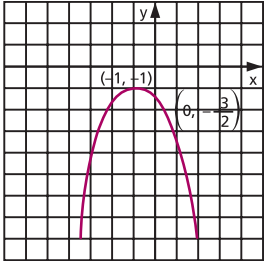
f)



g)



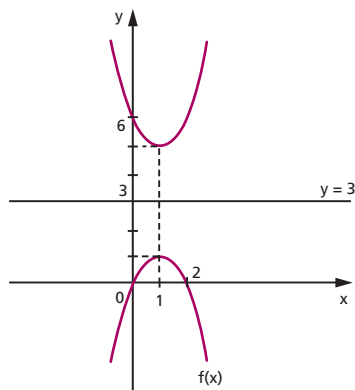
h)



282. $0 < c < b < a$

285. $a = 2$

286. $g(x) = x^2 - 2x + 6$



287. $A = 9$

288. a) $x^2 - 2x - 3 > 0 \Leftrightarrow x < -1$ ou $x > 3$
 $x^2 - 2x - 3 = 0 \Leftrightarrow x = -1$ ou $x = 3$
 $x^2 - 2x - 3 < 0 \Rightarrow -1 < x < 3$

b) $4x^2 - 10x + 4 > 0 \Leftrightarrow x < \frac{1}{2}$ ou $x > 2$
 $4x^2 - 10x + 4 = 0 \Leftrightarrow x = \frac{1}{2}$ ou $x = 2$

$4x^2 - 10x + 4 < 0 \Rightarrow \frac{1}{2} < x < 2$

c) $-x^2 + \frac{1}{2}x + \frac{1}{2} > 0 \Leftrightarrow -\frac{1}{2} < x < 1$
 $-x^2 + \frac{1}{2}x + \frac{1}{2} = 0 \Leftrightarrow x = -\frac{1}{2}$ ou $x = 1$
 $-x^2 + \frac{1}{2}x + \frac{1}{2} < 0 \Leftrightarrow x < -\frac{1}{2}$ ou $x > 1$

d) $-3x^2 + 6x - 3 = 0 \Leftrightarrow x = 1$
 $-3x^2 + 6x - 3 < 0 \Leftrightarrow x \neq 1$

e) $x^2 - 3x + \frac{9}{4} > 0 \Leftrightarrow x \neq \frac{3}{2}$
 $x^2 - 3x + \frac{9}{4} = 0 \Leftrightarrow x = \frac{3}{2}$

f) $3x^2 - 4x + 2 > 0, \forall x \in \mathbb{R}$

g) $-x^2 + x - 1 < 0, \forall x \in \mathbb{R}$

h) $-\frac{1}{2}x^2 - x - \frac{3}{2} < 0, \forall x \in \mathbb{R}$

289. $x_1 < x < x_2$ (x deve estar entre as raízes)

290. $\Delta < 0$ e $\begin{cases} a > 0 \Leftrightarrow f(x) > 0, \forall x \in \mathbb{R} \\ a < 0 \Leftrightarrow f(x) < 0, \forall x \in \mathbb{R} \end{cases}$

294. a) $S = \{x \in \mathbb{R} \mid x < 1 \text{ ou } x > 2\}$

b) $S = \{x \in \mathbb{R} \mid -2 < x < 3\}$

c) $S = \{x \in \mathbb{R} \mid x \leq -3 \text{ ou } x \geq \frac{1}{3}\}$

d) $S = \{x \in \mathbb{R} \mid -\frac{5}{2} \leq x \leq 4\}$

e) $S = \{x \in \mathbb{R} \mid \frac{1}{4} \leq x \leq \frac{3}{2}\}$

f) $S = \mathbb{R} - \left\{\frac{1}{2}\right\}$

g) $S = \mathbb{R}$

h) $S = \left\{\frac{3}{2}\right\}$

i) $S = \mathbb{R}$

j) $S = \mathbb{R}$

k) $S = \emptyset$

l) $S = \emptyset$

295. para todo x real

296. $A \cap B = \emptyset$

297. $\{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$

298. $20 < q(a) < 30$

299. $\{x \in \mathbb{R} \mid x \leq -2 \text{ ou } x \geq 2\}$

301. a) $S = \left\{x \in \mathbb{R} \mid -\frac{3}{2} < x < -\frac{1}{2} \text{ ou } 0 < x < \frac{1}{2}\right\}$

b) $S = \left\{x \in \mathbb{R} \mid 1 \leq x \leq \frac{3}{2} \text{ ou } 2 \leq x \leq \frac{5}{2}\right\}$

c) $S = \{x \in \mathbb{R} \mid -2 < x < 3 \text{ e } x \neq 1\}$

d) $S = \{x \in \mathbb{R} \mid x = -3 \text{ ou } 1 \leq x \leq 2\}$

e) $S = \{x \in \mathbb{R} \mid -1 < x < 1 \text{ ou } x > 2\}$

f) $S = \{x \in \mathbb{R} \mid x \leq 3\}$

302. a) $P_1(5, 0)$ e $P_2\left(-\frac{1}{2}, 0\right)$

b) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{2} \leq x \leq 5\right\}$

303. 19

304. $\{x \in \mathbb{R} \mid -4 < x < 2 \text{ ou } 3 < x < 4\}$

305. $A = \{x \in \mathbb{R} \mid 0 < x < 2 \text{ ou } 5 < x < 6\}$

307. a) $S = \left\{x \in \mathbb{R} \mid x < -\frac{5}{4} \text{ ou } -\frac{1}{2} < x < 1 \text{ ou } x > 2\right\}$

b) $S = \left\{x \in \mathbb{R} \mid x < -2 \text{ ou } -\frac{1}{3} < x \leq \frac{1}{3} \text{ ou } x \geq \frac{2}{3}\right\}$

c) $S = \{x \in \mathbb{R} \mid x < -3 \text{ ou } x \geq 0\}$

d) $S = \left\{x \in \mathbb{R} \mid -2 < x < \frac{1}{2} \text{ ou } x > \frac{2}{3}\right\}$

e) $S = \{x \in \mathbb{R} \mid -1 \leq x < 2 \text{ ou } 3 \leq x < 5\}$

f) $S = \left\{x \in \mathbb{R} \mid -2 < x < -\frac{3}{2} \text{ ou } -\frac{3}{4} < x < -\frac{1}{3}\right\}$

g) $S = \left\{x \in \mathbb{R} \mid -4 \leq x \leq -\frac{3}{4} \text{ ou } 1 < x < \frac{5}{2}\right\}$

h) $S = \{x \in \mathbb{R} \mid x > 0\}$

308. a) $S = \{x \in \mathbb{R} \mid -1 \leq x < 1 \text{ ou } x > 2\}$

b) $S = \{x \in \mathbb{R} \mid x \leq 0 \text{ ou } x > 1\}$

c) $S = \{x \in \mathbb{R} \mid x > 2\}$

d) $S = \{x \in \mathbb{R} \mid x < -1 \text{ ou } 0 \leq x < 1\}$

e) $S = \{x \in \mathbb{R} \mid t < 0\}$

f) $S = \{x \in \mathbb{R} \mid x < -1 \text{ ou } -1 < x \leq 0 \text{ ou } x > 1\}$

309. $\{x \in \mathbb{R} \mid x < 1\}$

310. $\{x \in \mathbb{R} \mid x \geq 3\}$

311. a) Significa obter para quais valores x a função está definida.

b) $D_f = \{x \in \mathbb{R} \mid -3 < x \leq -1 \text{ ou } 1 \leq x < 5\}$

312. $D_f = \{x \in \mathbb{R} \mid x < -3 \text{ ou } 2 < x \leq 5\}$

313. $\{x \in \mathbb{R} \mid x \geq 2\}$

314. $-4 \leq x < -3 \text{ ou } -1 < x \leq 2 \text{ ou } x \geq 3, x \in \mathbb{R}$

315. a) $S = \{x \in \mathbb{R} \mid 4 < x \leq 6\}$

b) $S = \{x \in \mathbb{R} \mid -3 \leq x < -2\}$

c) $S = \{x \in \mathbb{R} \mid -1 \leq x \leq 1 \text{ ou } 2 \leq x \leq 4\}$

d) $S = \{x \in \mathbb{R} \mid -3 \leq x < -1\}$

e) $S = \{x \in \mathbb{R} \mid -1 < x < 0\}$

f) $S = \emptyset$

316. a) $S = \{x \in \mathbb{R} \mid x < -2 \text{ ou } x > 3\}$

b) $S = \{x \in \mathbb{R} \mid -5 \leq x < -3\}$

c) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{2} \leq x < \frac{1}{2} \text{ ou } x > \frac{3}{2}\right\}$

d) $S = \left\{\frac{1}{2}\right\}$

317. a) V b) V c) F d) F e) V

318. a) F b) F c) F d) V e) V

320. a) $S = \{x \in \mathbb{R} \mid -3 \leq x \leq -1 \text{ ou } 1 \leq x \leq 3\}$

b) $S = \{x \in \mathbb{R} \mid x < -2 \text{ ou } x > 2\}$

c) $S = \{x \in \mathbb{R} \mid -1 < x < 1\}$

d) $S = \emptyset$

e) $S = \{x \in \mathbb{R} \mid x \leq -1 \text{ ou } x \geq 2\}$

f) $S = \mathbb{R}$

322. a) $m > \frac{9}{4}$ f) $1 < m < \frac{4}{3}$

b) $m \leq \frac{1}{4}$ g) $m \leq -2$

c) $0 < m < 4$ h) $m \geq 3$

d) $\nexists m \in \mathbb{R}$ i) $m < -2$

e) $\nexists m \in \mathbb{R}$ j) $m \geq 1$

324. a) $-2 < m < 2$ c) $m < -\frac{3}{4}$

b) $m < 1$ d) $-1 < m < 2$

325. $p > 11$

326. $-2 < m < -1$

327. $a > \frac{\sqrt{2}}{4}$

328. $0 < m < 8$

329. $k \geq 9$

331. a) $0 < m < \frac{7}{9}$

b) $m > 1$

c) $-2 < m < 0$

d) $-3 < m < x$

332. $m < -1$

333. $-2 < m < 2$

334. $k = 1$

336. $m < \frac{3}{2}$ ou $3 < m \leq \frac{7}{2}$

337. $m < -\frac{2\sqrt{2}}{3}$

338. $m < -5$

339. $-5 < m < -1$

340. $1 < m < 4$

341. $-\frac{3}{2} < m < -1$

342. $0 < m < \frac{1}{2}$

343. $m < \frac{3}{2}$ e $m \neq 0$ ou $m > 3$

344. $m > 1$

345. $-\sqrt{2} < m < -1$

346. $-1 < m < 2$

347. $m > 1$

348. $m < -2$ ou $2 < m < 3$

349. $-\frac{1}{4} \leq m < 0$ ou $m > 2$

350. $k = 1$

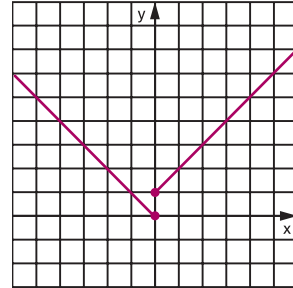
351. a) 49 b) 30 c) 6

352. $m < \frac{5}{2}$

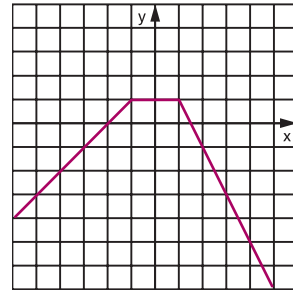
353. duas raízes negativas

Capítulo VIII

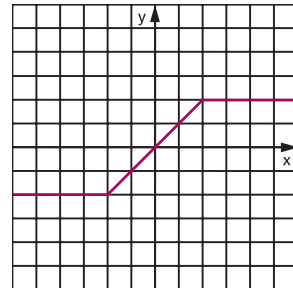
354. a)



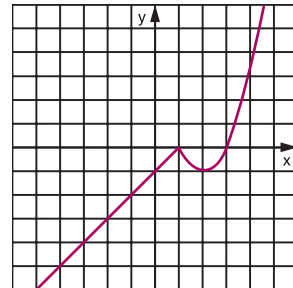
b)

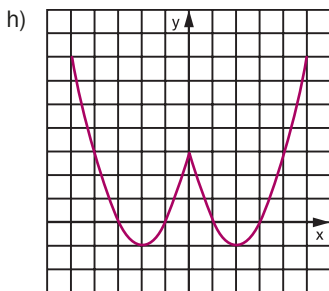
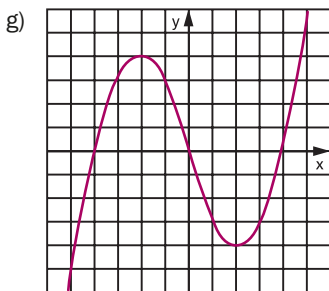
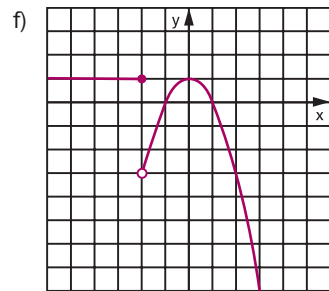
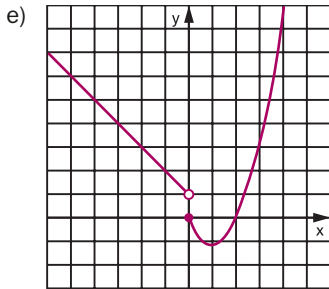


c)

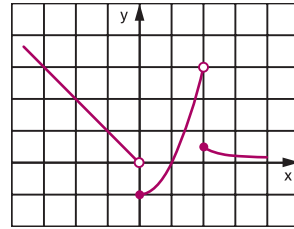


d)

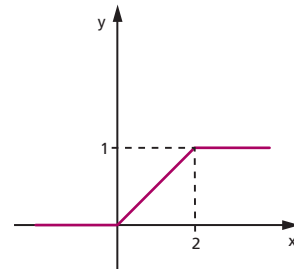




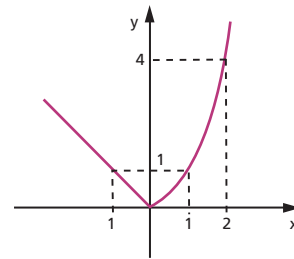
355.



356. a)

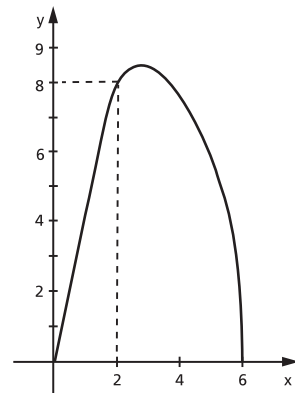


b)



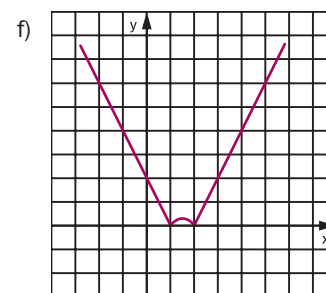
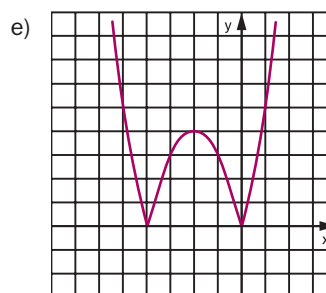
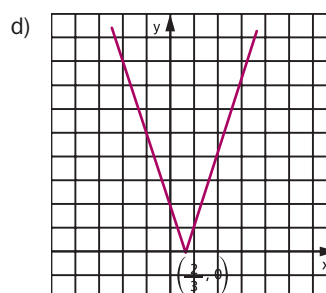
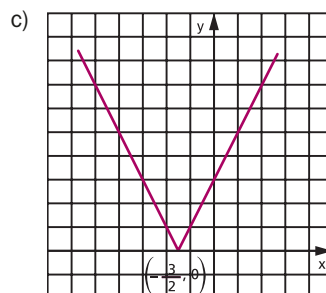
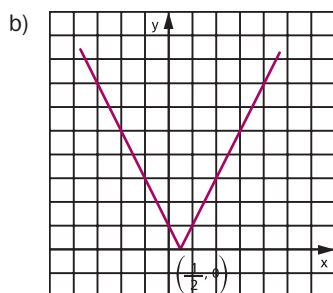
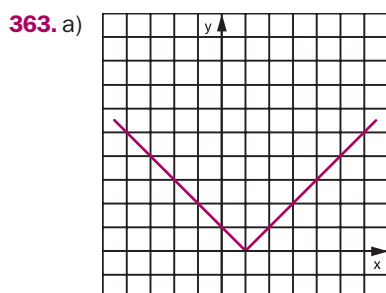
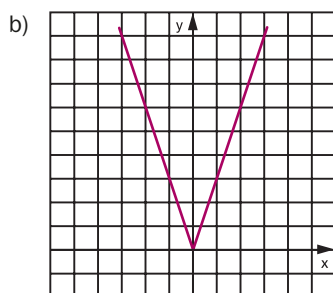
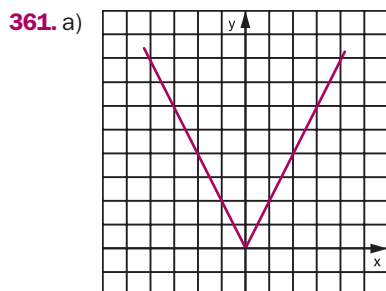
358. $x = 4$

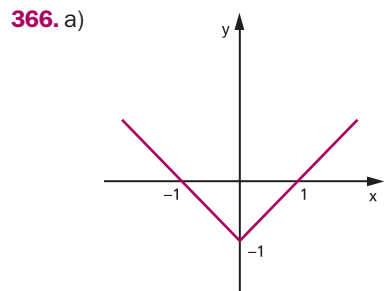
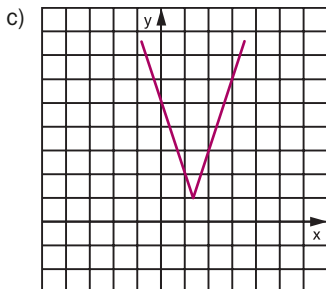
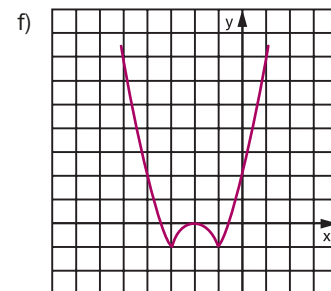
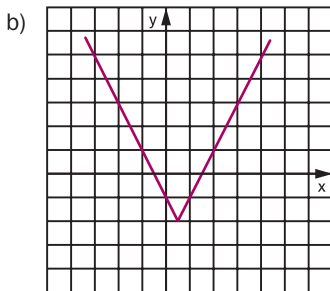
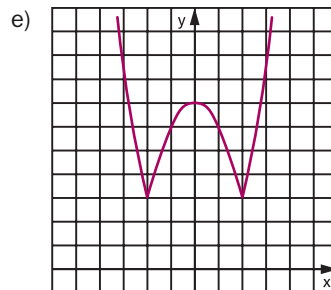
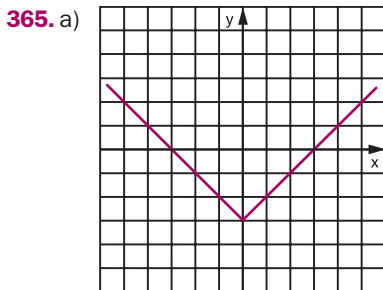
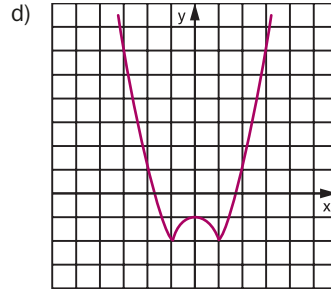
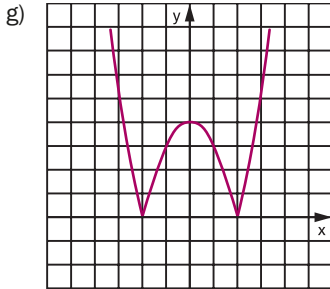
359. a)

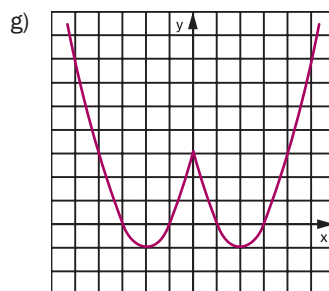
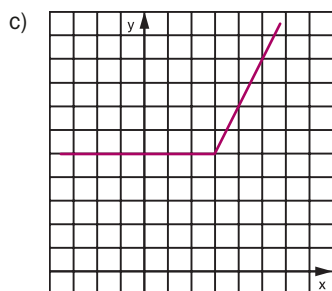
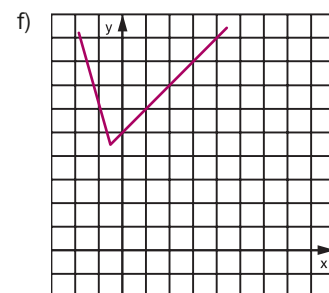
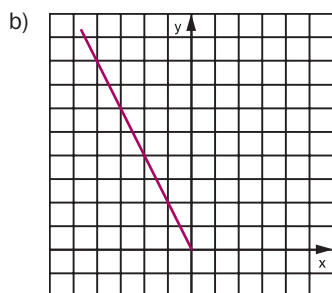
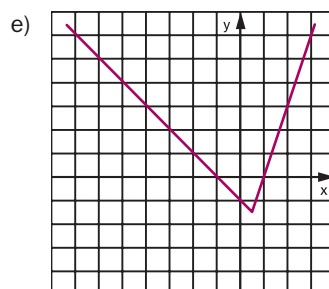
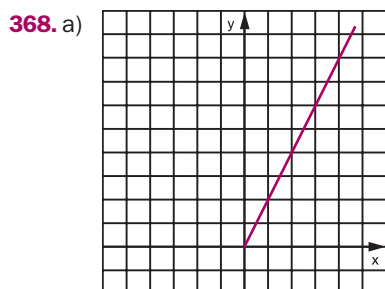
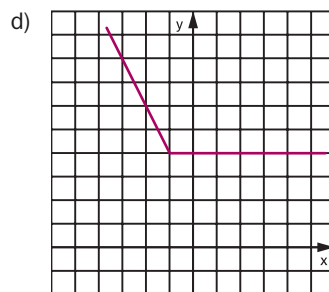
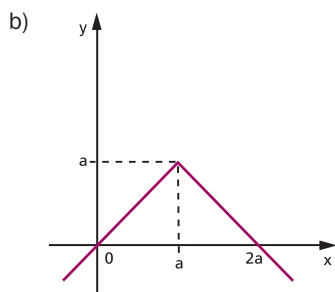


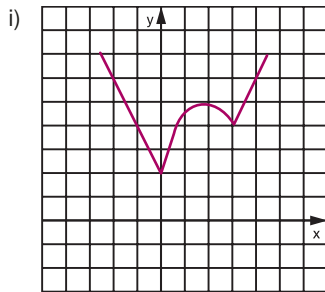
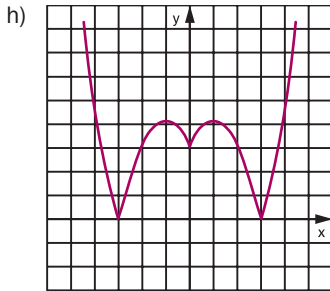
b) $f(x) = 5$ para $x = \frac{5}{4}$ ou $x = 5$

360. a) não c) sim e) sim
b) sim d) não

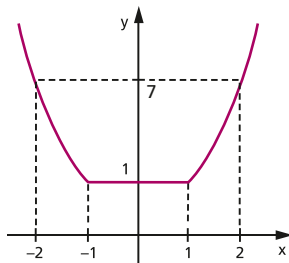








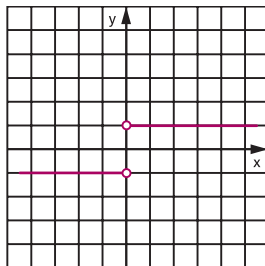
369.



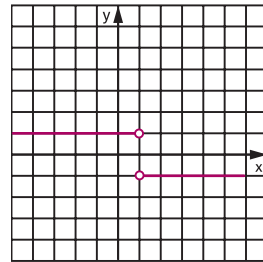
370. $\text{Im}_f = \{y \in \mathbb{R} \mid y \geq 2\}$

371. a, b, c

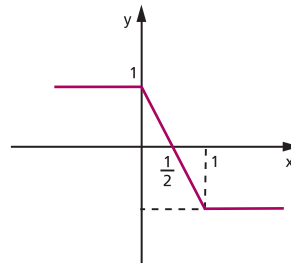
372.



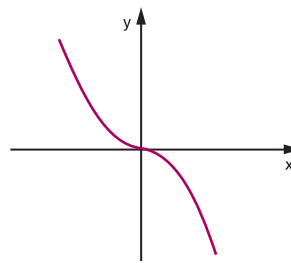
373.



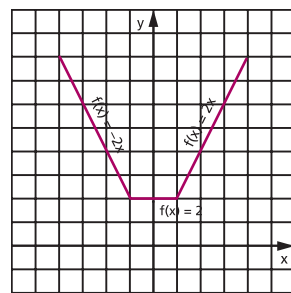
375. a)

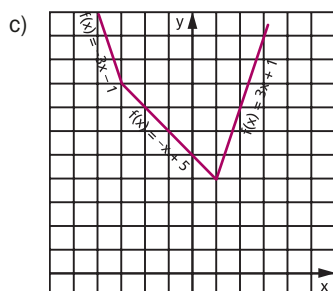
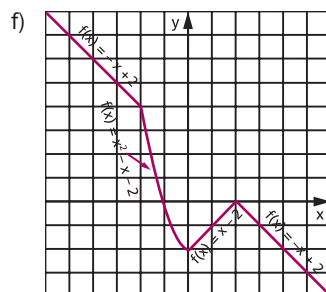
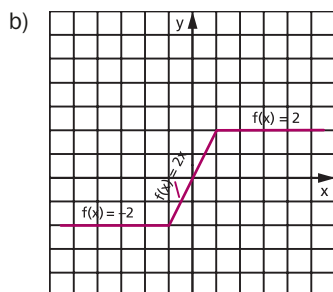


b)

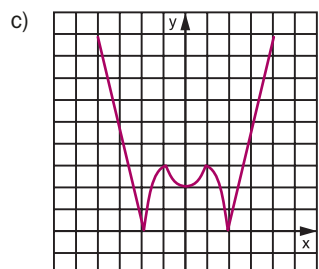
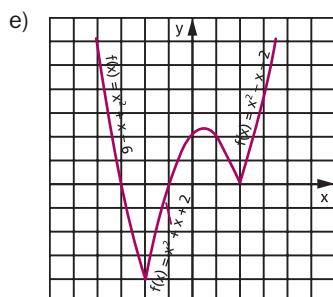
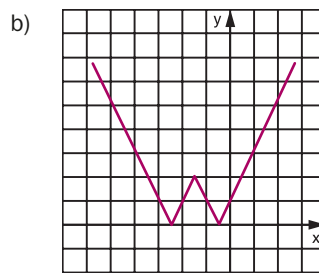
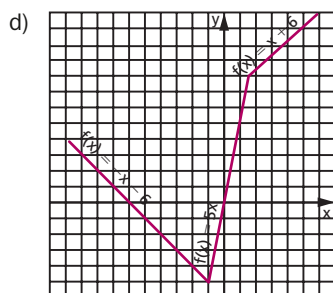
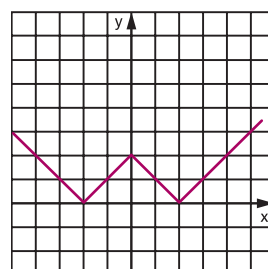


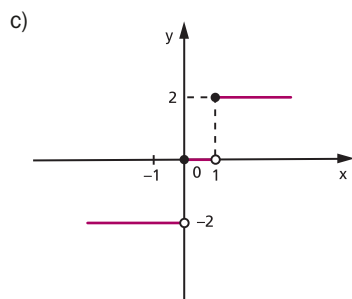
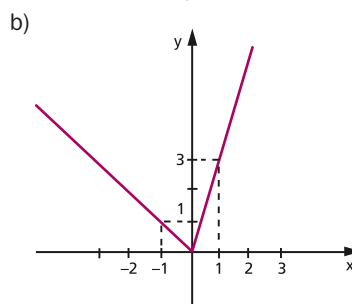
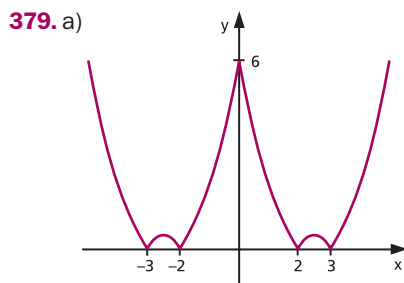
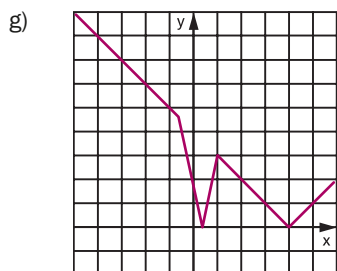
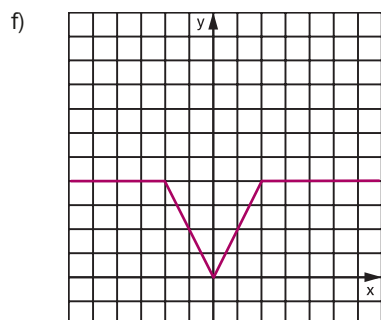
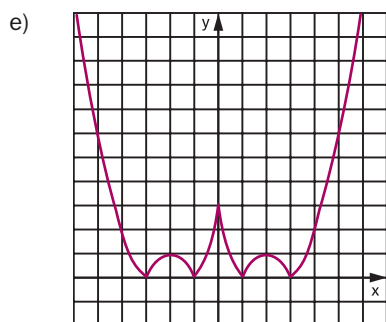
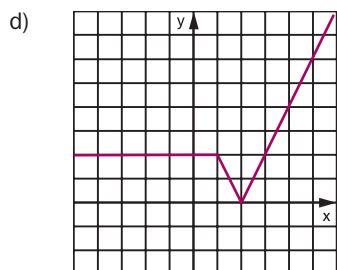
376. a)





378. a)





380. a, b, c e d

381. a) $S = \{1, -5\}$

b) $S = \left\{1, -\frac{1}{3}\right\}$

c) $S = \left\{\frac{5}{4}\right\}$

d) $S = \emptyset$

e) $S = \{-1, 1, 2, 4\}$

f) $S = \left\{-\frac{1}{2}, \frac{1}{2}, 2, 3\right\}$

g) $S = \{1, 3\}$

382. a) Demonstração

b) $k > 1$

383. a) $S = \left\{-\frac{3}{2}, -\frac{1}{4}\right\}$

b) $S = \left\{2, -\frac{1}{3}\right\}$

c) $S = \{-6, -1, 1, 4\}$

d) $S = \left\{-\frac{3}{2}, \frac{1}{3}, 1\right\}$

384. a) $S = \left\{\frac{1}{3}\right\}$

b) $S = \emptyset$

c) $S = \{4, 2\}$

d) $S = \{-13, -6\}$

e) $S = \left\{x \in \mathbb{R} \mid x \geq \frac{2}{3}\right\}$

f) $S = \left\{x \in \mathbb{R} \mid x \geq \frac{4}{3}\right\}$

385. $S = \{-2, 2\}$

387. a) $S = \{x \in \mathbb{R} \mid -1 \leq x \leq 0\}$

b) $S = \{x \in \mathbb{R} \mid x < 0 \text{ ou } x \geq 1\}$

388. a) $S = \left\{x \in \mathbb{R} \mid -\frac{2}{3} < x < 2\right\}$

b) $S = \{x \in \mathbb{R} \mid 1 \leq x \leq 2\}$

c) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{3} \leq x \leq 3\right\}$

d) $S = \left\{-\frac{4}{3}\right\}$

e) $S = \emptyset$

f) $S = \{x \in \mathbb{R} \mid x < -1 \text{ ou } x > 2\}$

g) $S = \left\{x \in \mathbb{R} \mid x \leq \frac{8}{5} \text{ ou } x \geq 0\right\}$

h) $S = \left\{x \in \mathbb{R} \mid x \leq \frac{1}{3} \text{ ou } x \geq 1\right\}$

i) $S = \left\{x \in \mathbb{R} \mid x \neq \frac{5}{3}\right\}$

j) $S = \mathbb{R}$

k) $S = \{x \in \mathbb{R} \mid -2 \leq x < 0 \text{ ou } 2 < x \leq 4\}$

389. a) $S = \{x \in \mathbb{R} \mid 1 < x < 2 \text{ ou } 3 < x < 4\}$

b) $S = \{x \in \mathbb{R} \mid x < -2 \text{ ou } -1 < x < 2 \text{ ou } x > 3\}$

c) $S = \{x \in \mathbb{R} \mid x \leq -1 \text{ ou } 2 \leq x \leq 3 \text{ ou } x \geq 6\}$

d) $S = \{x \in \mathbb{R} \mid -2 \leq x \leq 1 \text{ ou } 2 \leq x \leq 5\}$

e) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{4} < x < \frac{5}{8} \text{ e } x \neq \frac{1}{3}\right\}$

f) $S = \left\{x \in \mathbb{R} \mid x \leq \frac{1}{5} \text{ ou } x \geq 1\right\}$

g) $S = \{x \in \mathbb{R} \mid x < -3 \text{ ou } -1 < x < 1 \text{ ou } x > 3\}$

h) $S = \{x \in \mathbb{R} \mid x \leq -3 \text{ ou } -1 \leq x \leq 0 \text{ ou } x \geq 2\}$

i) $S = \{x \in \mathbb{R} \mid -3 \leq x \leq 0 \text{ ou } 1 \leq x \leq 4\}$

390. 29

391. a) V b) V c) V d) V e) F

392. 1

393. $S = \{x \in \mathbb{R} \mid -1 < x < 2 \text{ ou } 4 < x < 7\}$

394. $S = \{x \in \mathbb{R} \mid -2 < x < -1 \text{ ou } 0 < x < 1\}$

395. 3

396. a) V b) V c) F d) F e) V f) F

397. -1, 0, 3 e 4

399. a) $S = \{x \in \mathbb{R} \mid x \geq 3\}$

b) $S = \{x \in \mathbb{R} \mid x < 5\}$

c) $S = \{x \in \mathbb{R} \mid -1 \leq x \leq 1\}$

d) $S = \mathbb{R}$

e) $S = \emptyset$

f) $S = \{x \in \mathbb{R} \mid 3 \leq x \leq 6\}$

g) $S = \{x \in \mathbb{R} \mid 4 < x < 6\}$

400. $S = \{x \in \mathbb{R} \mid 1 < x < 4\}$

401. a, b, c, e

402. $S = \{x \in \mathbb{R} \mid x > 0\}$

404. a) $S = \{x \in \mathbb{R} \mid x < -5 \text{ ou } 1 < x < 5\}$

b) $S = \{x \in \mathbb{R} \mid x < -2 \text{ ou } x > 0\}$

c) $S = \{x \in \mathbb{R} \mid x \leq -5 \text{ ou } -3 \leq x \leq 7\}$

d) $S = \left\{x \in \mathbb{R} \mid -3 < x < \frac{11}{3}\right\}$

e) $S = \{x \in \mathbb{R} \mid x < -2 \text{ ou } x > 4\}$

f) $S = \{x \in \mathbb{R} \mid x \leq 0 \text{ ou } x \geq 3\}$

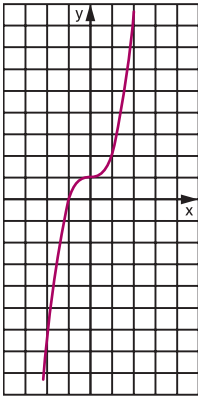
g) $S = \emptyset$

405. $S = \{x \in \mathbb{R} \mid x \leq 0 \text{ ou } x \geq 6\}$

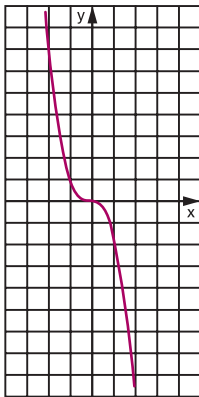
406. $S = \{x \in \mathbb{R} \mid x \geq -3\}$

Capítulo IX

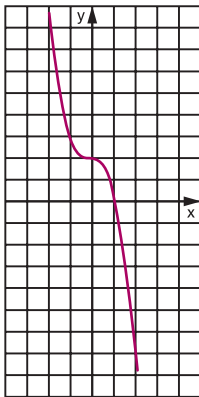
407. a)



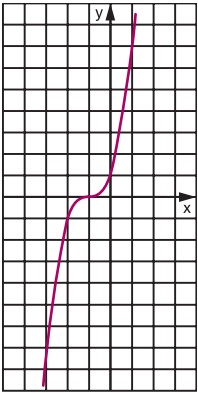
b)



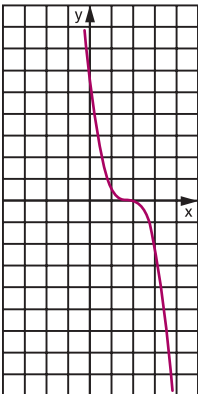
c)



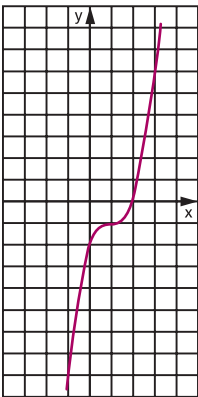
d)

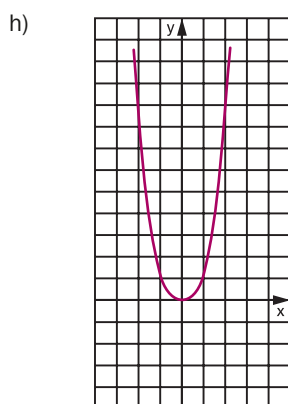


e)

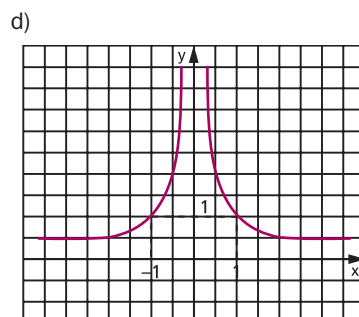
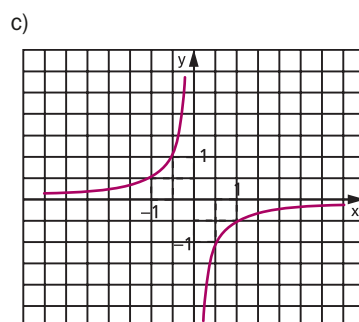
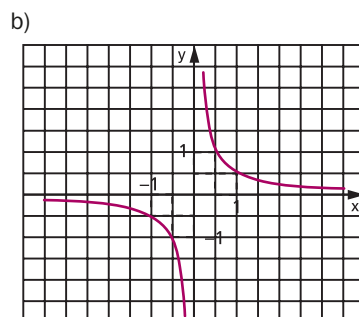
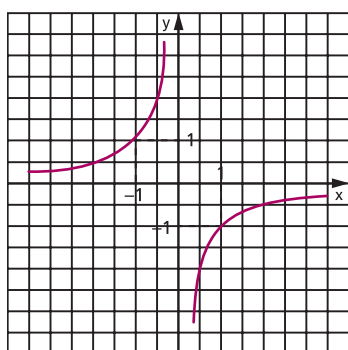


f)

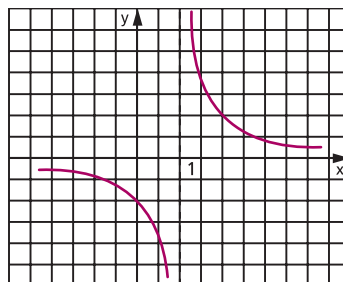




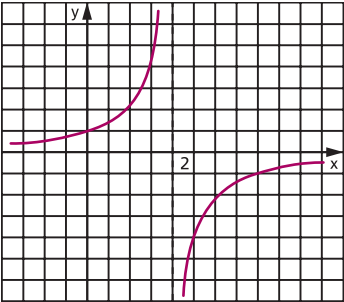
408. a)



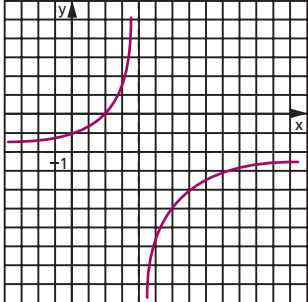
410. a)



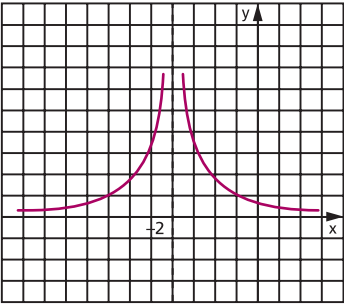
b)



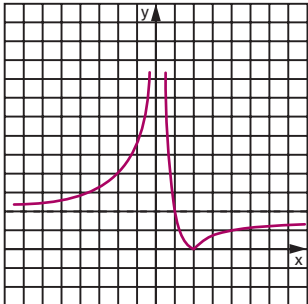
c)



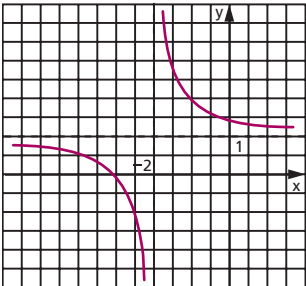
c)



d)

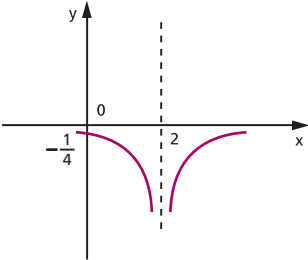


411. a)

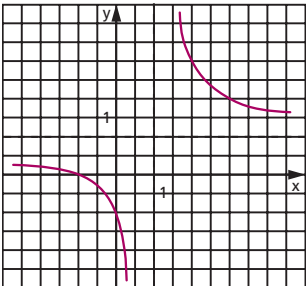


413. $\frac{35}{12}$

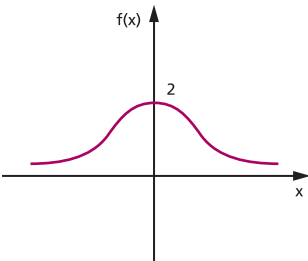
414. a)



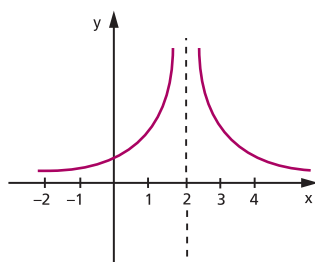
b)



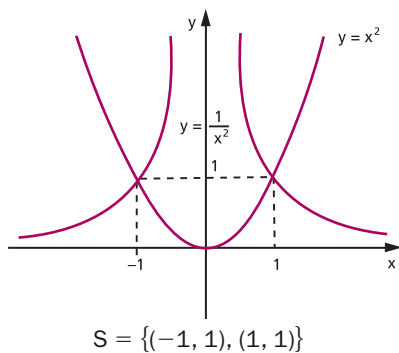
b)



c)

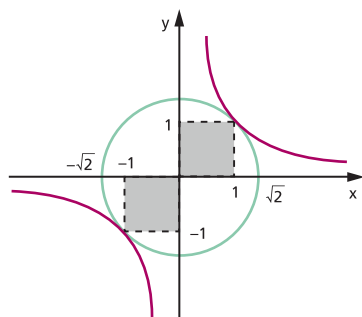


415.

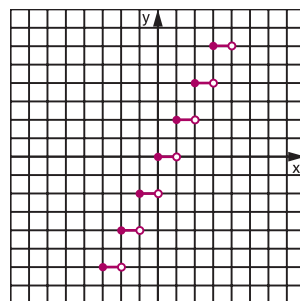


$$416. x = \frac{1 \pm \sqrt{5}}{2}$$

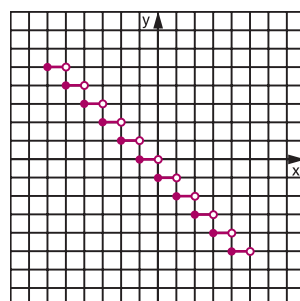
417.



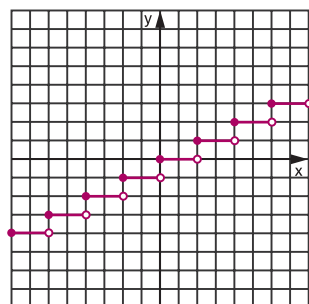
418. a)



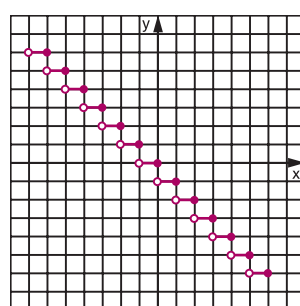
b)

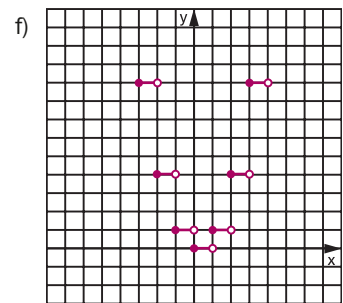
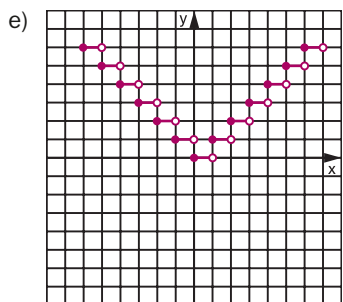
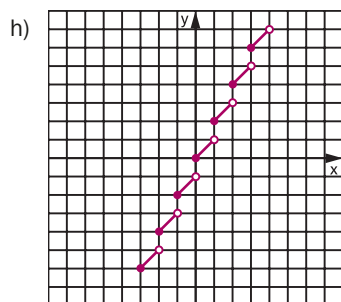
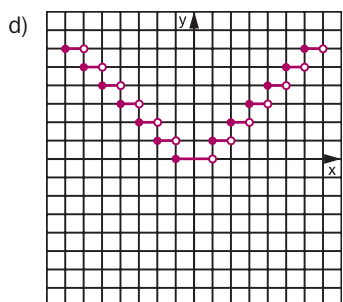
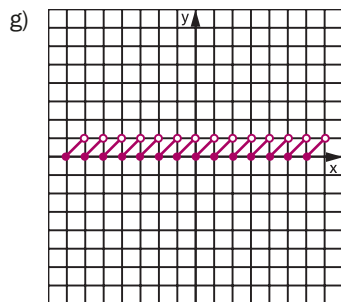
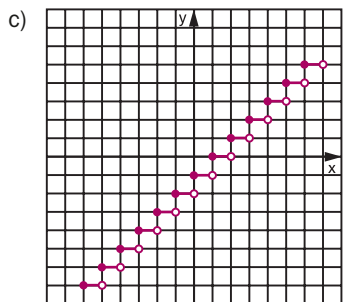


420. a)



b)





Capítulo X

422. a) $(f \circ g)(x) = 4x^2 - 2x - 2$

$(g \circ f)(x) = 5 + 2x - 2x^2$

b) $(f \circ g)(-2) = 18, (g \circ f)(-2) = -7$

c) $x = 2$ ou $x = -\frac{3}{2}$

423. $(f \circ g)(x) = x^4 - 6x^2 + 6$

$(g \circ f)(x) = x^4 - 8x^3 + 18x^2 - 8x$

424. $(f \circ g)(x) = 2, (g \circ f)(x) = 5$

425. a) $(f \circ g)(x) = x^2 - 6x + 11$

b) $(g \circ f)(x) = x^2 - 1$

c) $(f \circ f)(x) = x^4 + 4x^2 + 6$

d) $(g \circ g)(x) = x - 6$

426. $f(-x) = -x^3 - 3x^2 - 2x - 1$

$f\left(\frac{1}{x}\right) = \frac{1}{x^3} - \frac{3}{x^2} + \frac{2}{x} - 1$

$f(x - 1) = x^3 - 6x^2 + 11x - 7$

427. $a = 1$

431. a) $D(f \circ g) = \left\{ x \in \mathbb{R} \mid x \leq \frac{1}{2} \text{ ou } x \geq 2 \right\}$

b) $D(g \circ f) = \{x \in \mathbb{R} \mid x \geq 1\}$

432. a) $D(f \circ g) = \mathbb{R} - \left\{ -\frac{1}{2} \right\}$

$$(f \circ g)(x) = \frac{2x + 4}{2x + 1}$$

b) $D(g \circ f) = \mathbb{R} - \{2\}$

$$(g \circ f)(x) = \frac{5x - 4}{x - 2}$$

433. $[(h \circ g) \circ f](x) = 12x^2 + 12x + 2$

434. $[(h \circ (g \circ f))](x) = 2x^2 - 2x + 7$

435. $\theta = \pm \frac{\pi}{12} + k\pi, \theta = \frac{5\pi}{12} + k\pi$ ou

$$\theta = \frac{7\pi}{12} + k\pi$$

436. $(a, 3a - 3), \forall a \in \mathbb{R}$

437. $a = \frac{m + 4}{m + 2}$

438. a, c, e: falsos; b, d, f: verdadeiros

439. $f(g(x)) = 3$

440. $(f \circ [f \circ f])(x) = x$

441. $((h \circ f) \circ g)(2) = 5$

442. $x = -\frac{1}{2}$

443. $d(a - 1) = b(c - 1)$

445. $g(x) = \frac{x^2 - 2x - 4}{2}$

447. $f(x) = \frac{x^2 + 2x - 1}{2}$

448. $f(x) = \frac{2x + 4}{x - 1}$ para $x \neq 1$

449. $a = 1; b = \frac{1}{2}$

450. $b = -3$

451. $D_{f(x)} = \left\{ x \in \mathbb{R} \mid x \neq \frac{1}{2} \right\}$

452. $f\left(-\frac{12}{15}\right) = 7$

453. 7

454. a) $k = -1; t = 3$

b) $\left\{ x \in \mathbb{R} \mid x \leq \frac{1}{2} \text{ ou } x > 3 \right\}$

456. $(f \circ g)(x) = \begin{cases} 4x^2 + 4x & \text{se } x \geq -\frac{1}{2} \\ 4x + 3 & \text{se } x < -\frac{1}{2} \end{cases}$

$$(g \circ f)(x) = \begin{cases} 2x^2 - 8x + 9 & \text{se } x \geq 2 \\ 4x - 3 & \text{se } x < 2 \end{cases}$$

457. $(f \circ g)(x) = \begin{cases} 9x^2 - 12x + 6 & \text{se } x \geq 1 \\ -\frac{1}{3x} & \text{se } \frac{1}{3} < x < 1 \\ -9x^2 + 12x & \text{se } x \leq \frac{1}{3} \end{cases}$

$$(g \circ f)(x) = \begin{cases} -3x^2 - 4 & \text{se } x \leq -1 \\ \frac{2x - 7}{x - 2} & \text{se } -1 < x < 1 \\ 3x^2 - 10 & \text{se } x \geq 1 \end{cases}$$

458. $(f \circ g)(x) = \begin{cases} 4x + 1 & \text{se } x > 2 \\ 1 - 4x^2 & \text{se } -1 \leq x \leq 1 \\ x^4 + x^2 & \text{se } x < -1 \text{ ou } 1 < x \leq 2 \end{cases}$

$$(g \circ f)(x) = \begin{cases} 4x - 2 & \text{se } x > \frac{5}{4} \\ -16x^2 + 24x - 8 & \text{se } 0 \leq x \leq \frac{5}{4} \\ x^2 - 3x + 3 & \text{se } x < 0 \end{cases}$$

459. $f(x) = \begin{cases} x^2 + 3x - 1 & \text{se } x \geq -1 \\ 2x + 9 & \text{se } x < -1 \end{cases}$

460. a) injetora

b) sobrejetora

c) bijetora

d) não é injetora nem sobrejetora

461. a) injetora

b) bijetora

c) sobrejetora

d) não é injetora nem sobrejetora

462. a) III c) II e) II g) III

b) IV d) I f) III h) II

463. $b = 2$

464. $a = \frac{3}{4}$

465. $B = \{y \in \mathbb{R} \mid -2 \leq y < 3\}$

466. $B = \{y \in \mathbb{R} \mid 0 \leq y \leq 5\}$

467. a) III b) II c) I d) II e) II f) II

468. sobrejetora

469. a, b, e, f: falsas; c, d, g: verdadeiras

470. b, d, h: falsas; a, c, e, f, g: verdadeiras

471. a, c: falsas; b, d, e, f: verdadeiras

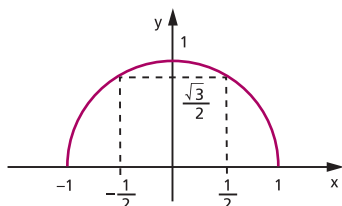
472. a, c, d: falsas; b, e: verdadeiras

473. a) $D_f = \{x \in \mathbb{R} \mid -1 \leq x \leq 1\}$

b) $\text{Im}_f = \{y \in B \mid 0 \leq y \leq 1\}$

c) Não, porque, por exemplo,
 $f\left(-\frac{1}{2}\right) = f\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$.

d)



474. a, b, d: falsas; c, e: verdadeiras

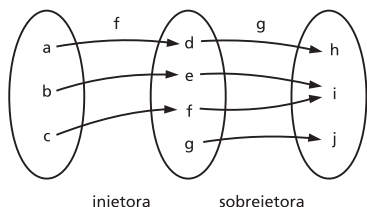
478. As funções I_A e I_B são iguais se, e somente se, $A = B$.

479. $m \leq n$, $m \geq n$, $m = n$

480. 12

481. 6

482.



$g \circ f$ não é injetora nem sobrejetora.

485. 8

486. a) $f^{-1}(x) = \frac{x-3}{2}$

b) $g^{-1}(x) = \frac{3x-1}{4}$

c) $h^{-1}(x) = \sqrt[3]{x-2}$

d) $p^{-1}(x) = 1 + \sqrt[3]{x-2}$

e) $q^{-1}(x) = x^3 - 2$

f) $r^{-1}(x) = x^3 + 1$

g) $s^{-1}(x) = \sqrt[3]{1-x^3}$

487. $f^{-1}(2) = 0$

488. $f^{-1}(x) = \sqrt[3]{x-1}$

489. Não, pois f não é injetora; por exemplo:
 $f(-1) = f(1) = 1$. Portanto f não é bijetora.

490. a, c, e: falsas; b, d: verdadeiras

492. a) $f^{-1}: \mathbb{R}_+ \rightarrow \mathbb{R}_+$

$f^{-1}(x) = \sqrt{x}$

b) $f^{-1}: \mathbb{R}_+ \rightarrow A$

$f^{-1}(x) = 1 - \sqrt{x}$

c) $f^{-1}: \mathbb{R}_- \rightarrow A$

$f^{-1}(x) = 2 - \sqrt{-x}$

d) $f^{-1}: \mathbb{R}_- \rightarrow A$

$f^{-1}(x) = -1 - \sqrt{-x}$

e) $f^{-1}: B \rightarrow \mathbb{R}_-$

$f^{-1}(x) = -\sqrt{x-1}$

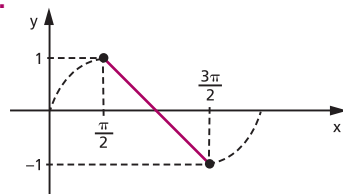
f) $f^{-1}: B \rightarrow \mathbb{R}_+$

$f^{-1}(x) = \sqrt{4-x}$

g) $f^{-1}: B \rightarrow \mathbb{R}_-$

$f^{-1}(x) = -\sqrt{x+1}$

493.



a, d, e: falsas; b, c: verdadeiras

494. b, f: não; a, c, d, e: sim

496. a) $f^{-1}: \mathbb{R} - \{1\} \rightarrow \mathbb{R} - \{3\}$

$$f^{-1}(x) = \frac{3x + 3}{x - 1}$$

b) $f^{-1}: \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{-1\}$

$$f^{-1}(x) = \frac{3 - x}{x - 2}$$

c) $f^{-1}: \mathbb{R} - \{-1\} \rightarrow \mathbb{R} - \{3\}$

$$f^{-1}(x) = \frac{3x + 4}{x + 1}$$

d) $f^{-1}: \mathbb{R} - \left\{\frac{5}{3}\right\} \rightarrow \mathbb{R} - \left\{\frac{1}{3}\right\}$

$$f^{-1}(x) = \frac{x + 2}{3x - 5}$$

e) $f^{-1}: \mathbb{R} - \{4\} \rightarrow \mathbb{R}^*$

$$f^{-1}(x) = \frac{2}{x - 4}$$

f) $f^{-1}: \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{3\}$

$$f^{-1}(x) = \frac{3x + 2}{x - 3}$$

497. $(f \circ g^{-1})(0) = 8$

498. $a = -1$

500. É o $\sqrt{17}$, pois $f^{-1}(\sqrt{17}) = 3$, isto é,
 $f(3) = \sqrt{17}$.

502. a) $f^{-1}: B \rightarrow A$

$$f^{-1}(x) = 1 + \sqrt{x + 1}$$

b) $f^{-1}: B \rightarrow A$

$$f^{-1}(x) = -1 + \sqrt{x - 1}$$

c) $f^{-1}: B \rightarrow A$

$$f^{-1}(x) = 2 - \sqrt{x + 1}$$

d) $f^{-1}: B \rightarrow A$

$$f^{-1}(x) = \frac{3 + \sqrt{4x + 1}}{2}$$

e) $f^{-1}: B \rightarrow A$

$$f^{-1}(x) = 2 + \sqrt{9 - x}$$

f) $f^{-1}: B \rightarrow A$

$$f^{-1}(x) = -1 - \sqrt{5 - x}$$

g) $f^{-1}: B \rightarrow A$

$$f^{-1}(x) = \frac{5 + \sqrt{8x + 9}}{4}$$

504. a)

$$f^{-1}(x) = \begin{cases} \frac{x - 3}{2} & \text{se } x \geq 7 \\ \frac{x - 1}{3} & \text{se } x < 7 \end{cases}$$

b)

$$f^{-1}(x) = \begin{cases} \frac{5 - x}{3} & \text{se } x \leq 8 \\ \frac{4 - x}{4} & \text{se } x > 8 \end{cases}$$

c)

$$f^{-1}(x) = \begin{cases} \sqrt{x} & \text{se } x \geq 0 \\ \frac{x}{2} & \text{se } x < 0 \end{cases}$$

d)

$$f^{-1}(x) = \begin{cases} \sqrt[3]{x + 2} & \text{se } x < -3 \\ \frac{x - 1}{4} & \text{se } x \geq -3 \end{cases}$$

e)

$$f^{-1}(x) = \begin{cases} x^2 + 3 & \text{se } x \geq 0 \\ 3 - \sqrt[3]{x} & \text{se } x < 0 \end{cases}$$

f)

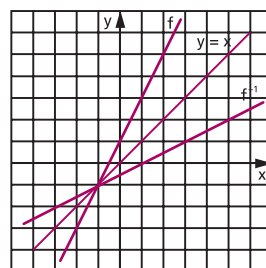
$$f^{-1}(x) = \begin{cases} 2 + \sqrt{x - 3} & \text{se } x \geq 3 \\ \frac{x + 1}{2} & \text{se } -3 < x < 3 \\ -1 - \sqrt{-x - 3} & \text{se } x \leq -3 \end{cases}$$

505. Não, pois f não é injetora, por exemplo:
 $f(-2) = f(1) = 3$. Portanto f não é bijetora.

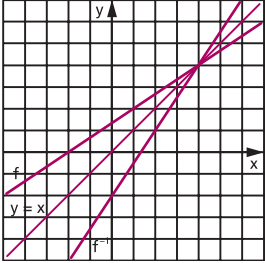
$$506. f^{-1}(x) = \begin{cases} x - 5 & \text{se } x \geq 7 \\ \frac{x + 3}{5} & \text{se } -8 \leq x < 7 \\ \frac{x + 5}{3} & \text{se } x < -8 \end{cases}$$

e $f^{-1}(42) = 37$

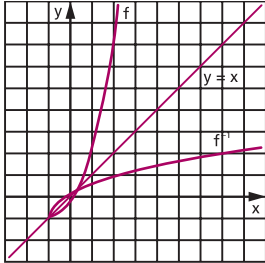
508. a)



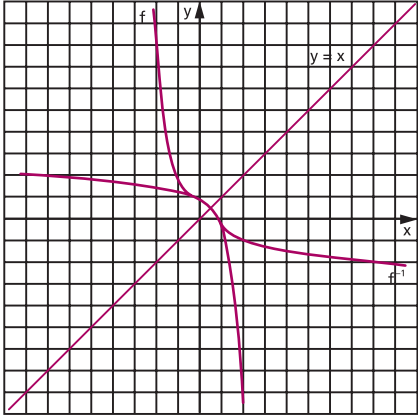
b)



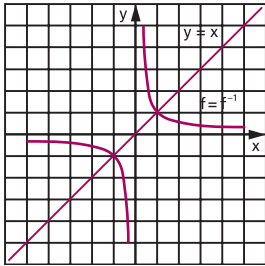
e)



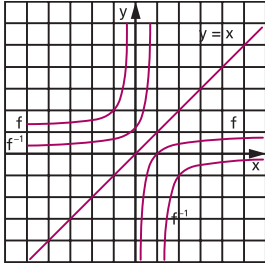
c)



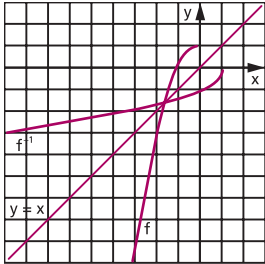
f)



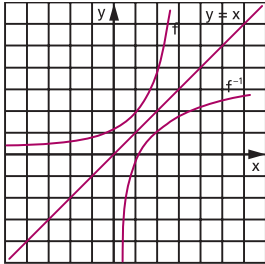
g)

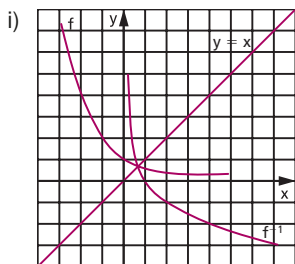


d)



h)





510. a) $(g \circ f)^{-1}: \mathbb{R} \rightarrow \mathbb{R}$

$$(g \circ f)^{-1}(x) = \frac{x+2}{12}$$

b) $(g \circ f)^{-1}: \mathbb{R} \rightarrow \mathbb{R}$

$$(g \circ f)^{-1}(x) = \sqrt[3]{\frac{x-3}{2}}$$

c) $(g \circ f)^{-1}: \mathbb{C} \rightarrow \mathbb{R}_+$

$$(g \circ f)^{-1}(x) = \sqrt{4-x}$$

d) $(g \circ f)^{-1}: \mathbb{R}_+ \rightarrow A$

$$(g \circ f)^{-1}(x) = \frac{3 + \sqrt{x}}{2}$$

e) $(g \circ f)^{-1}: \mathbb{C} \rightarrow A$

$$(g \circ f)^{-1}(x) = \sqrt{x^2 - 3}$$

511. Não, pois g não é injetora; por exemplo: $g(-1) = g(1) = 0$; portanto $g \circ f$ não é bijetora.

512. $[h \circ (g \circ f)]^{-1}: B \rightarrow A$

$$[h \circ (g \circ f)]^{-1}(x) = \frac{2 - \sqrt{x+1}}{4}$$

Apêndice I

514. a) $S = \{6\}$

b) $S = \{-4\}$

c) $S = \{1, 4\}$

d) $S = \left\{ \frac{7 + \sqrt{33}}{4}, \frac{7 - \sqrt{33}}{4} \right\}$

e) $S = \left\{ 0, \frac{7}{3} \right\}$

f) $S = \{77\}$

g) $S = \{13\}$

h) $S = \{3\}$

i) $S = \{3, 4\}$

j) $S = \{4\}$

k) $S = \{0\}$

l) $S = \{1\}$

m) $S = \emptyset$

n) $S = \left\{ 0, \frac{1}{4} \right\}$

o) $S = \left\{ \frac{5}{4} \right\}$

p) $S = \{0, 2\}$

515. $S = \{5\}$

516. $x = 2$

517. d, g: falsos; a, b, c, e, f: verdadeiros

518. $S = \emptyset$

520. a) $S = \{4, 9\}$

b) $S = \left\{ \frac{1}{9} \right\}$

c) $S = \emptyset$

d) $S = \{4 + 2\sqrt{3}\}$

e) $S = \{1, \sqrt[3]{25}\}$

f) $S = \{1\}$

g) $S = \{16\}$

h) $S = \{81\}$

i) $S = \left\{ 1, \frac{1}{16} \right\}$

j) $S = \left\{ 1, \frac{1}{16} \right\}$

522. a) $S = \left\{ -2, \frac{1}{3} \right\}$

b) $S = \{5, -1\}$

c) $S = \left\{ 4, -3, \frac{1 + \sqrt{29}}{2}, \frac{1 - \sqrt{29}}{2} \right\}$

d) $S = \emptyset$

e) $S = \left\{ 3, -\frac{5}{3} \right\}$

523. $S = \{0, 1, 4\}$

525. a) $S = \{64\}$

b) $S = \emptyset$

c) $S = \left\{\frac{5}{4}\right\}$

526. a) $S = \{0, 4\}$

b) $S = \{2\}$

c) $S = \{2, 6\}$

d) $S = \{1, 17\}$

527. a) $S = \{6\}$

b) $S = \left\{\frac{5}{11}\right\}$

c) $S = \{4\}$

529. a) $S = \{3\}$

b) $S = \{19\}$

530. a) $S = \{3\}$

b) $S = \left\{\frac{2}{3}\right\}$

c) $S = \{3, 4\}$

532. a) $S = \{1\}$

b) $S = \left\{\frac{1}{4}\right\}$

533. a) $S = \{1\}$

b) $S = \left\{\frac{25}{9}\right\}$

534. $\begin{cases} a < b \Rightarrow S = \{a\} \\ a \geq b \Rightarrow S = \{b\} \end{cases}$

535. $S = \left\{\frac{3a}{4}\right\}$

536. $\begin{cases} a = b = 0 \Rightarrow S = \mathbb{R}_+ \\ a \geq b > 0 \Rightarrow S = \left\{\frac{(a-b)^2}{4b}\right\} \\ a < b \text{ ou } b = 0 \Rightarrow S = \emptyset \end{cases}$

537. a) $b \geq 1 \Rightarrow S = \left\{\frac{2a\sqrt{b}}{b+1}\right\}$

b) $a = b \Rightarrow S = \{x \in \mathbb{R}_+ \mid x \geq a\}$
 $a \neq b \Rightarrow S = \{a + b\}$

d) $S = \{34\}$

e) $S = \left\{\frac{177}{4}\right\}$

f) $S = \{40\}$

e) $S = \{8\}$

f) $S = \emptyset$

g) $S = \left\{\frac{4}{\sqrt{5}}, -\frac{4}{\sqrt{5}}\right\}$

d) $S = \{3\}$

e) $S = \{3a\}$

c) $S = \{2\}$

d) $S = \{3\}$

d) $S = \{4\}$

e) $S = \{3\}$

c) $S = \{2\}$

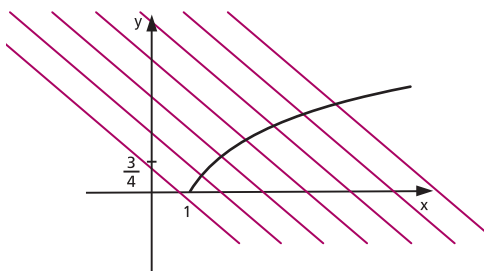
c) $S = \{0\}$

c) $b > a \Rightarrow S = \left\{\frac{2a^2b}{a^2 + b^2}\right\}$

538. $a < 0$ e $|b| \geq |a| \rightarrow S = \left\{0, \frac{5a^2 - b^2}{4a}\right\}$

539. $a \geq \frac{3}{4}; x = \frac{(2a+1) - \sqrt{4a-3}}{2}$

porque a interseção se dá nos pontos de menor abscissa.



540. a) $S = \{(9, 4), (4, 9)\}$

b) $S = \{(10 + 4\sqrt{6}, 10 - 4\sqrt{6})\}$

c) $S = \{(2, 8), (8, 2)\}$

d) $S = \{(9, 4), (4, 9)\}$

541. a) $S = \left\{(4, 2), \left(-\frac{9}{2}, \frac{41}{12}\right)\right\}$

b) $\{(-4, 6)\}$

543. a) $S = \{2\}$

b) $S = \left\{\frac{7}{4}\right\}$

c) $S = \{-16\}$

d) $S = \{4, -3\}$

e) $S = \left\{-\frac{2}{3}, 3\right\}$

f) $S = \{4 + \sqrt{3}, 4 - \sqrt{3}\}$

g) $S = \{0\}$

h) $S = \left\{0, \frac{3 + \sqrt{3}}{4}, \frac{3 - \sqrt{3}}{4}\right\}$

i) $S = \left\{0, \frac{1}{4}, \frac{3}{2}\right\}$

j) $S = \left\{0, -3, \frac{1}{4}\right\}$

544. $S = \emptyset$

546. $S = \{-2, 7\}$

547. $x^2 = 80$

548. $S = \left\{1, \frac{3}{2}, 2\right\}$

549. $S = \{1, 2, 10\}$

551. a) $S = \{0\}$

b) $S = \left\{\frac{\sqrt{5}}{2}, -\frac{\sqrt{5}}{2}\right\}$

c) $S = \left\{\frac{4}{5}\right\}$

552. $S = \{(8, 64), (64, 8)\}$

$$3 \leq x < 12\}$$

i) $S = \left\{x \in \mathbb{R} \mid \frac{1 + \sqrt{13}}{6} \leq x \leq 1 \text{ ou } x \geq 2\right\}$

556. $S = \left\{x \in \mathbb{R} \mid x < \frac{3 - \sqrt{41}}{16}\right\}$

558. a) $S = \{x \in \mathbb{R} \mid x > 11\}$

b) $S = \{x \in \mathbb{R} \mid x \geq -2\}$

c) $S = \left\{x \in \mathbb{R} \mid x \geq \frac{3}{4}\right\}$

d) $S = \left\{x \in \mathbb{R} \mid x < \frac{1}{4} \text{ ou } x > 3\right\}$

e) $S = \{x \in \mathbb{R} \mid x \leq 1 - \sqrt{3} \text{ ou } x \geq 1 + \sqrt{3}\}$

f) $S = \left\{x \in \mathbb{R} \mid -4 \leq x \leq \frac{1}{5}\right\}$

g) $S = \emptyset$

Apêndice II

554. a) $S = \left\{x \in \mathbb{R} \mid \frac{2}{3} \leq x < 2\right\}$

b) $S = \left\{x \in \mathbb{R} \mid -\frac{5}{2} \leq x \leq 2\right\}$

c) $S = \{x \in \mathbb{R} \mid -2 < x \leq -1 \text{ ou } 2 \leq x < 3\}$

d) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{3} \leq x \leq \frac{2}{3} \text{ ou } 1 \leq x \leq 2\right\}$

e) $S = \emptyset$

555. a) $S = \left\{x \in \mathbb{R} \mid 1 \leq x \leq \frac{4}{3}\right\}$

b) $S = \{x \in \mathbb{R} \mid x > 4\}$

c) $S = \{x \in \mathbb{R} \mid x > 8\}$

d) $S = \{x \in \mathbb{R} \mid x \geq 1\}$

e) $S = \left\{x \in \mathbb{R} \mid -1 \leq x < \frac{7 - \sqrt{17}}{2}\right\}$

f) $S = \{x \in \mathbb{R} \mid 2 \leq x \leq 3\}$

g) $S = \{x \in \mathbb{R} \mid x > 1\}$

h) $S = \left\{x \in \mathbb{R} \mid -1 < x \leq -\frac{1}{2} \text{ ou } 1 \leq x < 2\right\}$

559. a) $S = \{x \in \mathbb{R} \mid 1 < x < 2\}$

b) $S = \{x \in \mathbb{R} \mid x \leq 2\}$

c) $S = \{x \in \mathbb{R} \mid x \geq 2 - \sqrt{6}\}$

d) $S = \left\{x \in \mathbb{R} \mid x < -\frac{1}{2} \text{ ou } x > 2\right\}$

e) $S = \{x \in \mathbb{R} \mid x \leq 1\}$

f) $S = \left\{x \in \mathbb{R} \mid x \leq -2 - 2\sqrt{2} \text{ ou } -2 + 2\sqrt{2} \leq x \leq \frac{6 + 2\sqrt{3}}{3}\right\}$

g) $S = \emptyset$

h) $S = \mathbb{R}$

i) $S = \{x \in \mathbb{R} \mid -1 \leq x \leq 2\}$

j) $S = \left\{x \in \mathbb{R} \mid -\frac{1}{2} \leq x < 2\right\}$

561. a) $S = \left\{x \in \mathbb{R} \mid -\frac{3}{5} \leq x < 0 \text{ ou } x > 3\right\}$

b) $S = \{x \in \mathbb{R} \mid -6 \leq x < 0 \text{ ou } 3 < x \leq 4\}$

c) $S = \{x \in \mathbb{R} \mid 0 < x \leq 2\}$

d) $S = \left\{x \in \mathbb{R} \mid \frac{3}{2} \leq x \leq 2\right\}$

563. a) $S = \left\{ x \in \mathbb{R} \mid x \geq \frac{3}{2} \right\}$

b) $S = \left\{ x \in \mathbb{R} \mid -\frac{2}{3} < x \leq 5 \right\}$

c) $S = \left\{ x \in \mathbb{R} \mid 3 \leq x \leq \frac{13 + \sqrt{201}}{4} \right\}$

d) $S = \left\{ x \in \mathbb{R} \mid -2 \leq x \leq \frac{3}{2} \text{ ou } 3 \leq x \leq 4 \right\}$

e) $S = \left\{ x \in \mathbb{R} \mid x < 2 - \sqrt{3} \text{ ou } x \geq 3 + \sqrt{2} \right\}$

f) $S = \{x \in \mathbb{R} \mid 2 \leq x \leq 3\}$

g) $S = \emptyset$

h) $S = \mathbb{R}$

564. a) $S = \left\{ x \in \mathbb{R} \mid -2 < x < -\frac{5 + \sqrt{13}}{2} \right\}$

b) $S = \left\{ x \in \mathbb{R} \mid \frac{-3 - \sqrt{5}}{2} < x \leq 1 \right\}$

c) $S = \{x \in \mathbb{R} \mid -1 \leq x \leq 1\}$

d) $S = \{x \in \mathbb{R} \mid x > 1\}$

566. a) $S = \{x \in \mathbb{R} \mid x > 11\}$

b) $S = \{x \in \mathbb{R} \mid x \geq 4\}$

c) $S = \left\{ x \in \mathbb{R} \mid -1 \leq x < 1 - \frac{\sqrt{31}}{8} \right\}$

d) $S = \left\{ x \in \mathbb{R} \mid x \leq -2 \text{ ou } -1 \leq x < \frac{-1 + \sqrt{13}}{6} \right\}$

567. $S = \left\{ x \in \mathbb{R} \mid \frac{5}{2} \leq x < 3 \right\}$

568. $S = \left\{ x \in \mathbb{R} \mid -\frac{45}{4} < x < 1 \text{ ou } x \geq 9 \right\}$